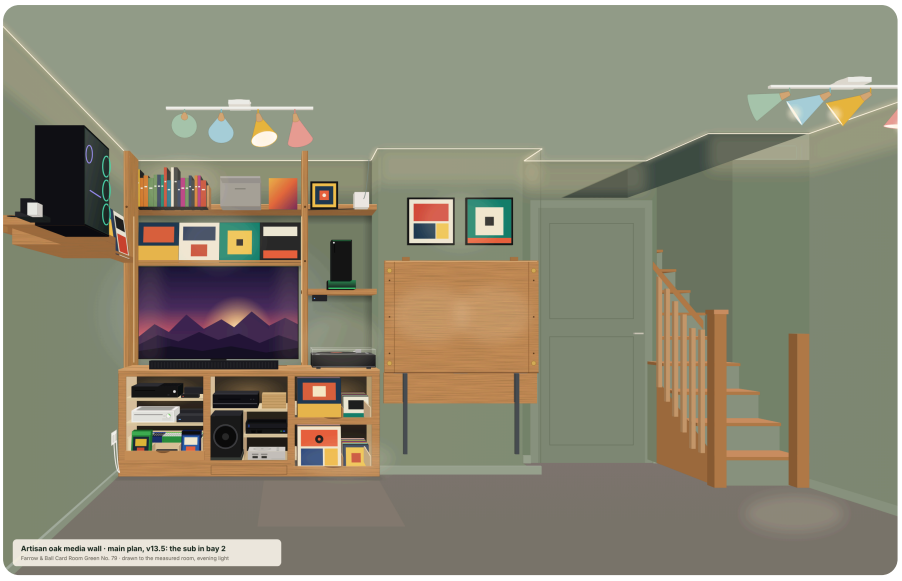
Home Network & Media Wall Plan

Revised Oct 20, 2026 · @emma · version 14 (revision 14.1: a faster 3D room, on GitHub Pages), drawn to your measurements

What changed in v13. You asked for less cladding all over the place, so it's gone: no slatted cable spine, no slatted PC screen, no slat panel behind the turntable, no slatted sub box, no reeded table. The PC's leads now run in conduit **chased into the solid side wall**, so there's no trunking either; the sub stands under a **plain oak cover** (below); the folded table is a **plain oiled oak panel** between its pegged breadboard ends. The oak that's left is the joinery that does a job: the uprights, the shelves, the ledges, the worktop, the face frame and the table. The **stairs go up** to the first floor (drawn rising away from the room; tell us if they run the other way). New: **the drawer layouts**, every drawer to scale with what goes in it (section 3). The slat-wall alternative is retired.
v13.1–13.2: the sub gets a **plain oak cover** with no sides: spills run off it, the sub can't tip, and nothing blocks its driver or port (section 5). A new render shows **every drawer out**, and the viewer can lay **rough measurements** over any view. The **stairwell opening** now starts just after the door, over the banister, and the sofa is drawn as yours. **Six at the table** now use the leaf too: 170 × 80, 85 cm a diner on the long sides and 75 cm off the breast, every clearance checked. The **side view** now shows how deep the alcove is round the wall: 38 of the 46 cm base inside it, 8 proud, the uprights and shelves wholly inside.
v13.3: the build guide is now step by step, each step with what to buy before it and what to check after it; the buy list shows what each phase costs; and the network and power section lists what each part does, both setup stages and what plugs into every socket (section 9).
v13.5: the sub is back in the media wall. The two Xbox bricks lie flat at the back of their shelves by the power leads, the RetroTINK and the HDMI switch move into the space that frees in bay 1, and the sub takes the lower left of bay 2, under the TV, on an isolation riser beside the PS2 and PS1 (section 3). The dining wall has nothing under it now, so eight can sit down without moving anything.
v13.7: four **lighting scenes** (TV night, dinner, records, everyday evening), each drawn on the room plan with every light's level, so you know where to set each dimmer (section 8).
v13.8: the whole plan is back on your own deck (the one you can edit, with your saved answers), and the summaries now follow the fit proof and the questions list, so they can't fall out of date: the PC case is answered and off the list, and the three checks on site are the TV's plugs, the RetroTINK's size and the sub's vibration.
v13.6: the **leaf carries the legs**: a slim oak frame underneath holds all eight oak legs in clips, so the folded table stays a plain panel, nothing new stands in the room, and the leaf goes wherever the chairs are kept (section 5). **The electrics are your own job**: swapping the socket for a double, the two dimmers and the two spot bars are like-for-like and not notifiable in England, so there's no electrician's visit; there is a safe routine and a clear point to stop and call one (section 9). The PC now runs from the 10-gang. Every step and every buy-list phase carries a **coloured label** for its part (Network, Media wall, Dining table and so on), and one list says what we still need from you.
v14: a **second place for the table**: on the wall by the stairs, under your 2 m shelf (section 5 compares the two). The plan page now has a **3D room** you can walk round: switch the table's wall, fold it or set it for four or six, change the paint, the lighting scene and where the floor lamp stands, and save any view as an HD picture.

1 · The plan on a page



Drawn to your measurements, in Card Room Green: the alcove with the turntable zone on the right, the dining wall folded on the chimney breast, both spot bars. Full-resolution renders come with this plan.

£2,764–5,317
the whole plan, item by item in section 11; no electrician on top

49 of 66
fit checks pass outright; 13 are tight, 4 need a check on site (section 3)

188 · 240 · 4–8
LPs and singles, all face-out · seats at the table

Area	Decision
Alcove	The TV between two oak uprights, the left one now a corner post on the side wall; on the right, the turntable on the worktop and the Series X on a shelf at 134; below, three 60 cm bays: the Xbox consoles, the PlayStations with the sub, and the two record trays.
Side wall	The record ledge wraps the corner with four more LPs, then an oak shelf for the PC above the end of the sofa, clear of the TV's sightlines.
Dining wall	A 110 × 80 solid oak table on an oak backboard: folded flat every day, down on two Häfele Hebgo brackets for four, lifted off with its leaf for six or eight. The leaf carries all eight oak legs in a slim frame underneath.
Records	Every record face-out: about 188 LPs (180 in the trays, 8 on the two ledges) and 240 singles.
Heights	Worktop and table 76 · TV centre 116.5 · Series X shelf 134 · ledges and PC shelf 155.4 · high shelf 196, 39 under the 235 ceiling.
Colour and light	Card Room Green No. 79, colour-drenched. The ceiling LED strip, LED grooves in the uprights, a bias light, and two Wolston-style spot bars on your two existing ceiling points: between the TV and the sofa, and between the door and the wall.
Power and network	Your own job: the side-wall socket becomes a double, left for the 10-gang (everything modern, the PC included), right for the retro set, switched at the wall. Deco A in bedroom 3, Cat6 down the airing cupboard.

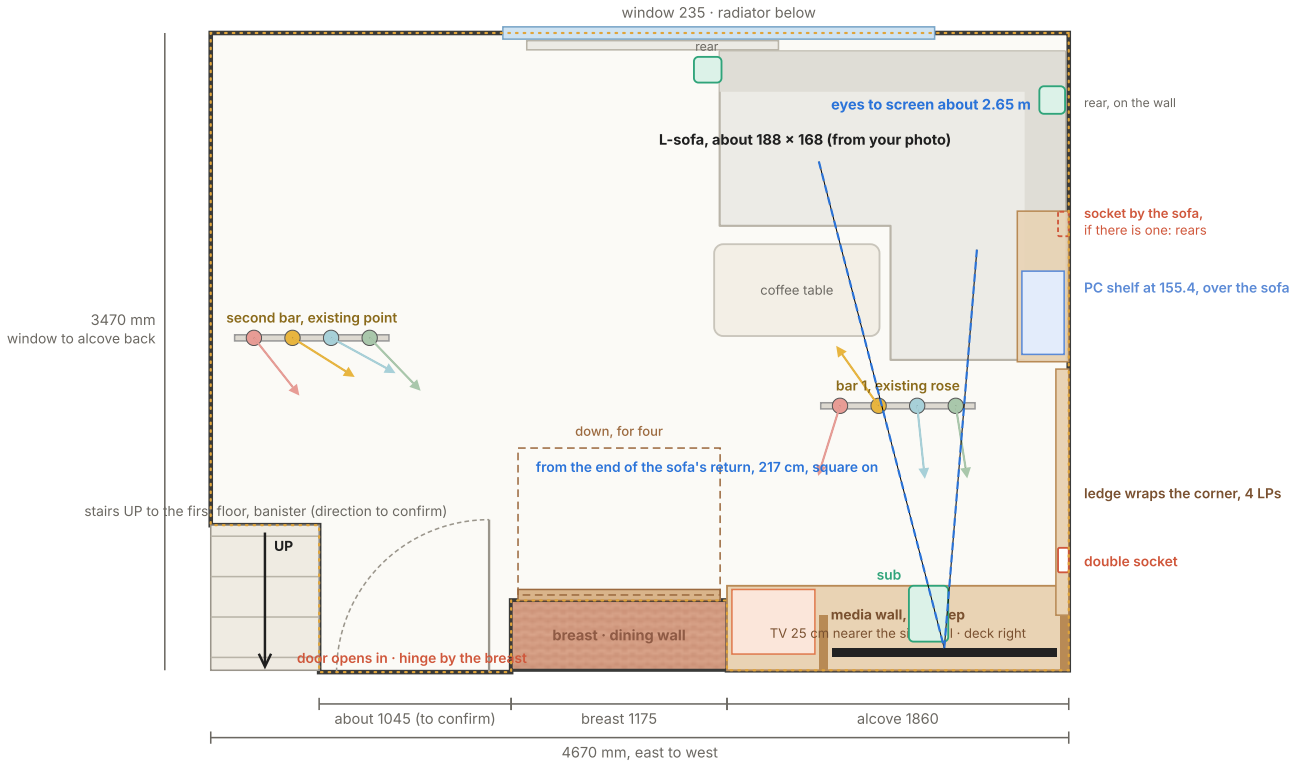
To confirm (section 12): which dining spot · the stairs wall's build · the floor lamp's spot · that you're swapping the light fittings yourselves · the sofa's return length · the stairs wall's sizes · the turntable's model · the fan stand's model · chairs · the 23 × 22 cm box by the window · record counts · game counts · extra wired ports · the 10-gang · console model numbers · the pendant point · the RetroTINK's size.

2 · The room

Everything is drawn from your hand-measured sketch; where the agent's floor plan disagrees, your tape wins. The L-sofa is drawn from your photo[24]: its long side ends in line with the breast's reveal (about 188 cm); the return along the side wall is a typical 168 until you measure it.

The living room from above, as you measured it

To scale, north at the top. Media wall in the alcove, the ledge and PC shelf on the side wall over the sofa, the dining wall on the chimney breast, two spot bars



Ceiling 2350 mm. Sofa from your photo (long side to the reveal; return 168 assumed). The stairs go up; their direction is drawn roughly.



New in v14: the whole room in 3D, from the window on an everyday evening, the table folded on the breast. On the plan page you can walk round it, switch the table's wall, the paint, the lights and the lamp's spot, and save any view in HD.

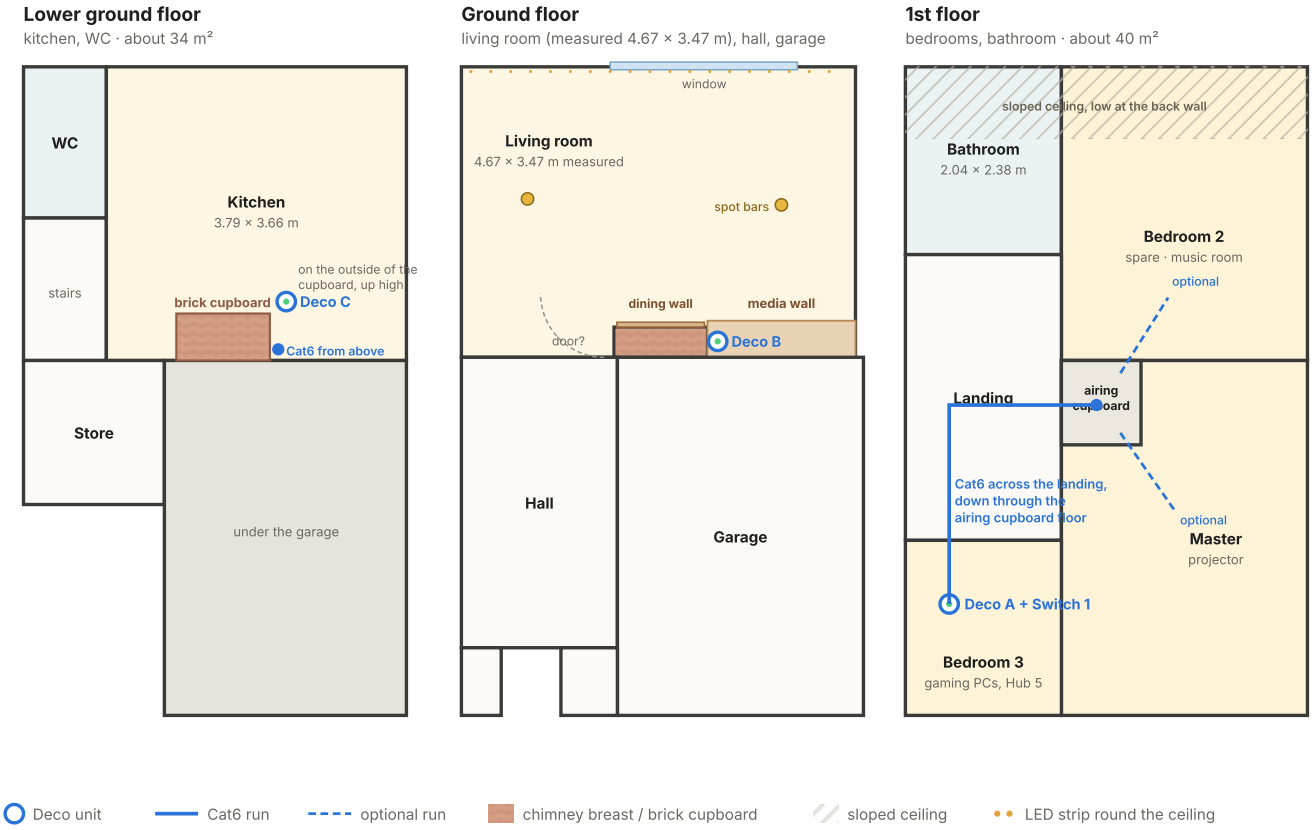
From your sketch	mm	Notes
Room, east to west along the window	4670	The media wall is the south wall, shared with the hall and garage
Window wall to the alcove's back wall	3470	To the chimney breast's face: 3090
Ceiling	2350	
TV alcove, east wall to the breast	1860 × 380	Floor to ceiling
Chimney breast	1175	Projects 380 on the alcove side, 390 on the west side: the dining wall
Window and radiator (north wall)	2350 · 1370	Window starts 730 from the east wall; radiator ends 130 inside its west end
Skirting	60 × 20	White now; Card Room Green eggshell in the plan
West end: recess, door, stair box	935 / 1045 / 660 / 820	Drawn roughly: please confirm which is which
Stairs, beside the door	820 wide	They go **up** to the first floor (you confirmed), behind a wooden banister; drawn rising away from the room until you send a photo

From the sofa: from the middle of the long side your eyes are about 2.65 m from the screen, which a 55-inch fills at about 26°: right for TV and games. With the TV 25.25 cm nearer the side wall, it now sits square on to the end of the sofa's return (217 cm away), and only about 6° off-centre from the middle of the long side. The screen's centre at 116.5 cm is 1–2.5° above a seated eye line. Both spot bars hang 22.4 cm, so the lowest shade is at 212.6 cm: no headroom worries.

The house

The house: where the Decos and Cat6 runs go

Deco A and Switch 1 move to bedroom 3 with the Hub 5. Both Cat6 runs drop through the airing cupboard, right over the media wall's corner



Deco unit · Cat6 run · optional run · chimney breast / brick cupboard · sloped ceiling · LED strip round the ceiling

Outline traced from the agent's plan (north, the window side, at the top). The living room's alcove, chimney breast and windows are placed from your measurements.

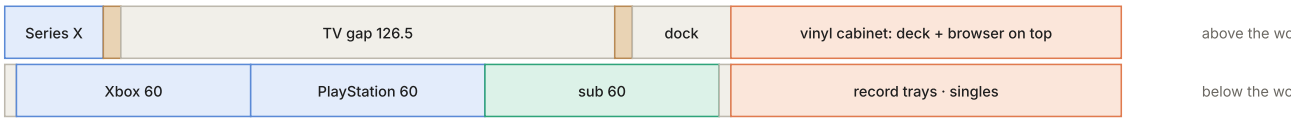
Deco A and Switch 1 sit with the Hub 5 and the gaming PCs in bedroom 3; both Cat6 runs go across the landing and drop through the airing cupboard, which sits over the alcove's corner (section 9).

3 · The media wall

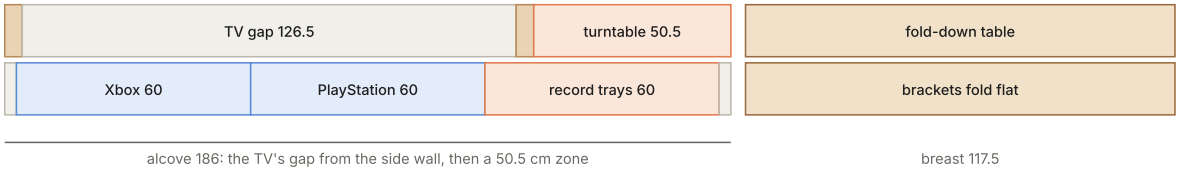
What moved in v11

To scale, from the side wall. Top: v10. Bottom: v11

v10: the TV in the middle, the vinyl cabinet on the breast



v11: the TV 25.25 cm nearer the side wall, the dining wall on the breast



- Moving the TV 25.25 cm left joins the two 25.25 cm side gaps into one 50.5 cm zone: wide enough for a 45.2 cm turntable, with 2.65 cm each side.
- The Series X moves up onto a shelf over the deck, above the lid's swing, with 15 cm of air each side. The records sit in bay 3, right under the deck.
- That frees the chimney breast for a fold-down dining table; the sub, which is wireless, moves to the floor by the side wall.

Why this layout, and what we ruled out

- **The TV in the middle (v10):** left two 25.25 cm gaps either side, too narrow for a turntable, so the vinyl had to move out to the chimney breast. Moving the TV joins them into one 50.5 cm zone.
- **The turntable with the Series X beside it:** 45.2 + 20.5 cm won't go in 50.5. The Series X goes up on a shelf instead, above the lid's swing, where it gets more air than it had in v10 (15 cm each side, 21 above).
- **The vinyl cabinet on the breast as well as a table:** a table needs the breast's full width, and a cabinet in front would put the diners 44 cm further into the room.
- **A gateleg table in the recess:** it would block the door; the breast is the only clear wall.



Numbered view: the game drawer and the lower record tray are pulled out 30 cm, with a view of each from above. Dashed lines show what's hidden: gold for data, audio and low-voltage leads, coral for mains. The sofa isn't drawn so the base stays in view; the room plan shows it.

OAK JOINERY	
1	Solid oak uprights, 4.5 × 30, worktop to ceiling (159 long)
2	The left upright is a corner post against the side wall
3	Flush wedged tenons carry the Series X shelf (134)
4	High shelf, top 196: books up to 30 cm
5	Record ledge above the TV: 4 LPs face-out
6	The ledge wraps the corner: 4 more LPs face-out
7	LED groove in each upright washes the wall
8	High shelf, right: the “now playing” 7-inch
ON THE WORKTOP	
9	LG OLED C6 55in · centre 116.5 · now 25 cm nearer the side wall
10	Soundbar · eARC, optical and AUX inputs
11	Turntable (AT-LP120X size allowed) · lid clear to 130
12	Xbox Series X on its fan stand · 15 cm air each side
13	Phono preamp, hung under the shelf at the back
14	Solid oak worktop 186 × 46 × 3, top at 76
IN THE ALCOVE BASE	
15	Oak face frame; its bottom rail hides the 9 cm plinth
16	Xbox One + HDMI switch · its brick flat at the back
17	Xbox 360 + RetroTINK · its brick flat at the back
18	Game drawer: 2 face-out bins, spine-up rows behind
19	PS3 + controller basket · bay 2 top row
20	PS2 · bay 2, beside the sub
21	PS1 · under the PS2
22	Upper record tray: ~90 LPs + ~120 singles · bay 3
23	Lower record tray: the same, shown pulled out
24	Hidden slim drawer in the plinth, finger pull
25	Sub · bay 2, under the TV, on an isolation riser

THE DINING WALL ON THE CHIMNEY BREAST	
26	Oak backboard: lift-off hinges on top, brackets below
27	Oak table 110 × 80: folded, a plain oiled oak panel
28	Häfele Hebgo folding brackets, 500 kg a pair
29	Turn buttons hold the folded table
30	Two framed LPs above
THE SIDE WALL	
31	PC on its oak shelf, above the end of the sofa (see the room plan)
32	Brush grommet: the PC's leads drop into the wall
33	Pad charging dock
34	PC cables chased into the wall: no spine, no trunking
35	Keyboard tray under the shelf, rolls out (v6.1)
POWER, NETWORK AND LIGHT	
36	Deco B: leads in the groove behind the right upright
37	Double socket: its right switch is the retro switch
38	10-gang extension, hidden behind the TV
39	LED strip round the ceiling edge (as now)
40	Spot bar on the existing rose, between TV and sofa
41	Second spot bar, existing point by the door
42	Stairs UP to the first floor, beside the door (direction to confirm)

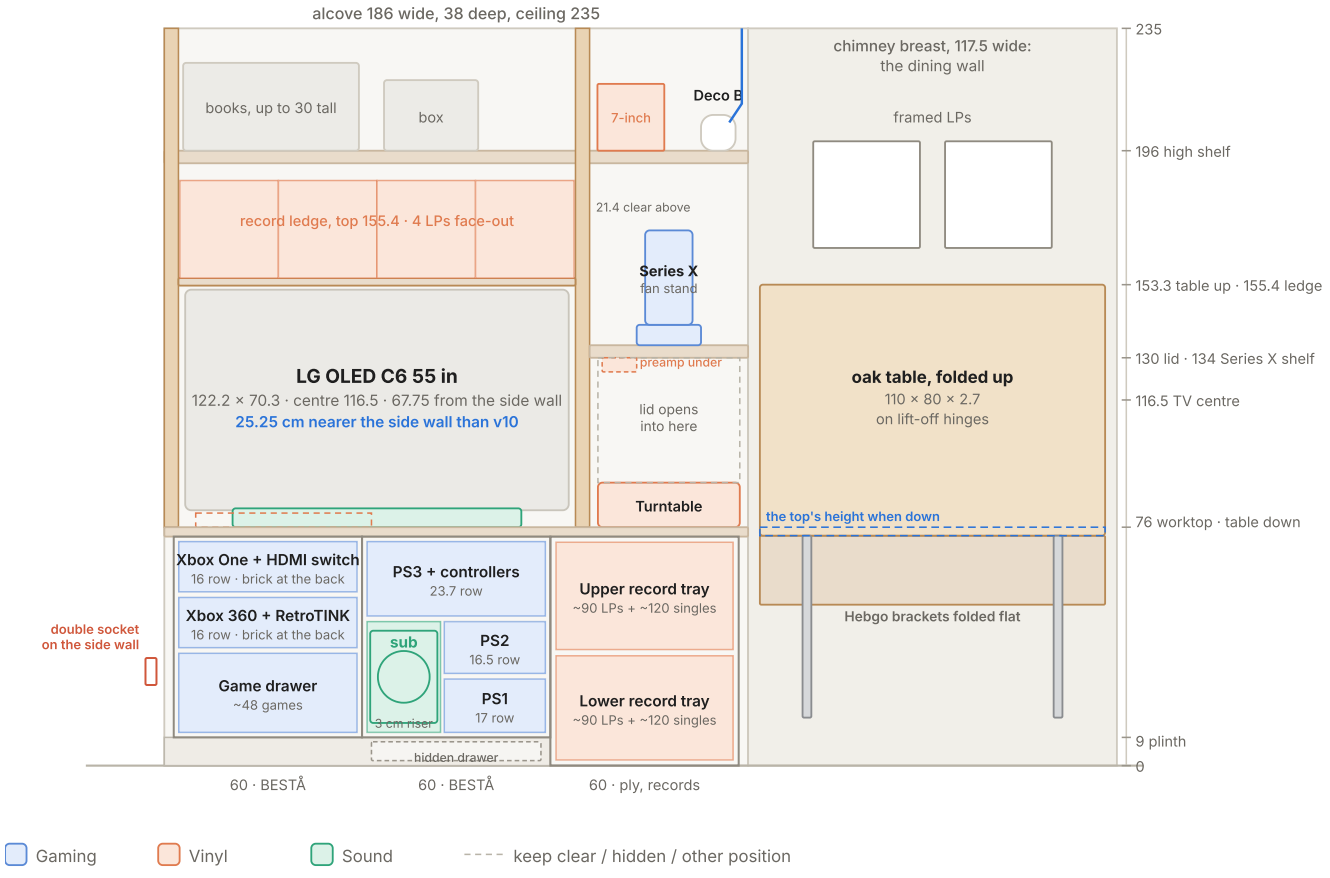
Built to your alcove: 186 wide, 38 deep, 235 to the ceiling

- **Base:** 3 cm filler, three 60 cm bays, 3 cm filler. Bays 1 and 2 are IKEA BESTÅ 60×40×64 frames on a 9 cm plinth (tops at 73, back panels off); bay 3 is an 18 mm ply carcass 60 × 40 × 73 from the floor, for the record trays. One oak face frame runs across all three; its bottom rail hides the plinth and a hidden drawer under bay 2 with a finger pull (no push-to-open at floor level).
- **Depth:** 40 cm frames, a 4 cm cable channel behind and the 2 cm face frame make 46 cm from the back wall, 8 cm proud of the chimney breast.
- **Worktop:** solid oak, 186 × 46 × 3, top at 76, with a brush-lined slot behind the TV and a grommet behind the deck.

- **Oak uprights:** 4.5 × 30 cm, 159 cm from the worktop to the ceiling. The left one stands against the side wall at 0–4.5, a corner post that frames the TV and carries the wrap ledge. The right one is at 131–135.5. That leaves 126.5 cm between them, 2.15 cm clear of the TV each side.
- **TV:** centre 67.75 cm from the side wall (it was 93), on LG's stand set 2 cm off the back wall: bottom 81.4, centre 116.5, top 151.7. The soundbar sits centred under it at the worktop's front edge.
- **The turntable zone:** 50.5 cm between the right upright and the breast. The deck sits on the worktop with 2.65 cm each side and its lid opens to about 121. Above it is an oak shelf at 134, through-tenoned into the upright, carrying the Series X on its fan stand; the phono preamp hangs underneath at the back.
- **Shelves above:** the record ledge at 155.4 between the uprights, four LPs face-out; the high shelf at 196, wall to wall in two lengths, with books, a box, a print, the framed "now playing" 7-inch and Deco B.
- **Bay 2 with the sub (v13.5):** the PS3 keeps the top row. Below its shelf the bay splits: on the left the sub's column, 23.4 cm wide, then a 12 mm ply divider, then 32.2 cm for the PS2 over the PS1 on a part shelf. The sub stands on a 3 cm isolation riser with 1 cm each side and 2.9 above. Its front is flush with the face frame, so its driver faces the room and its rear port has 15.6 cm to the wall.
- **Power:** the side-wall socket in front of bay 1 becomes a double. Its two leads enter through a notch at the left end of the face frame's bottom rail and run back through the plinth to the channel.

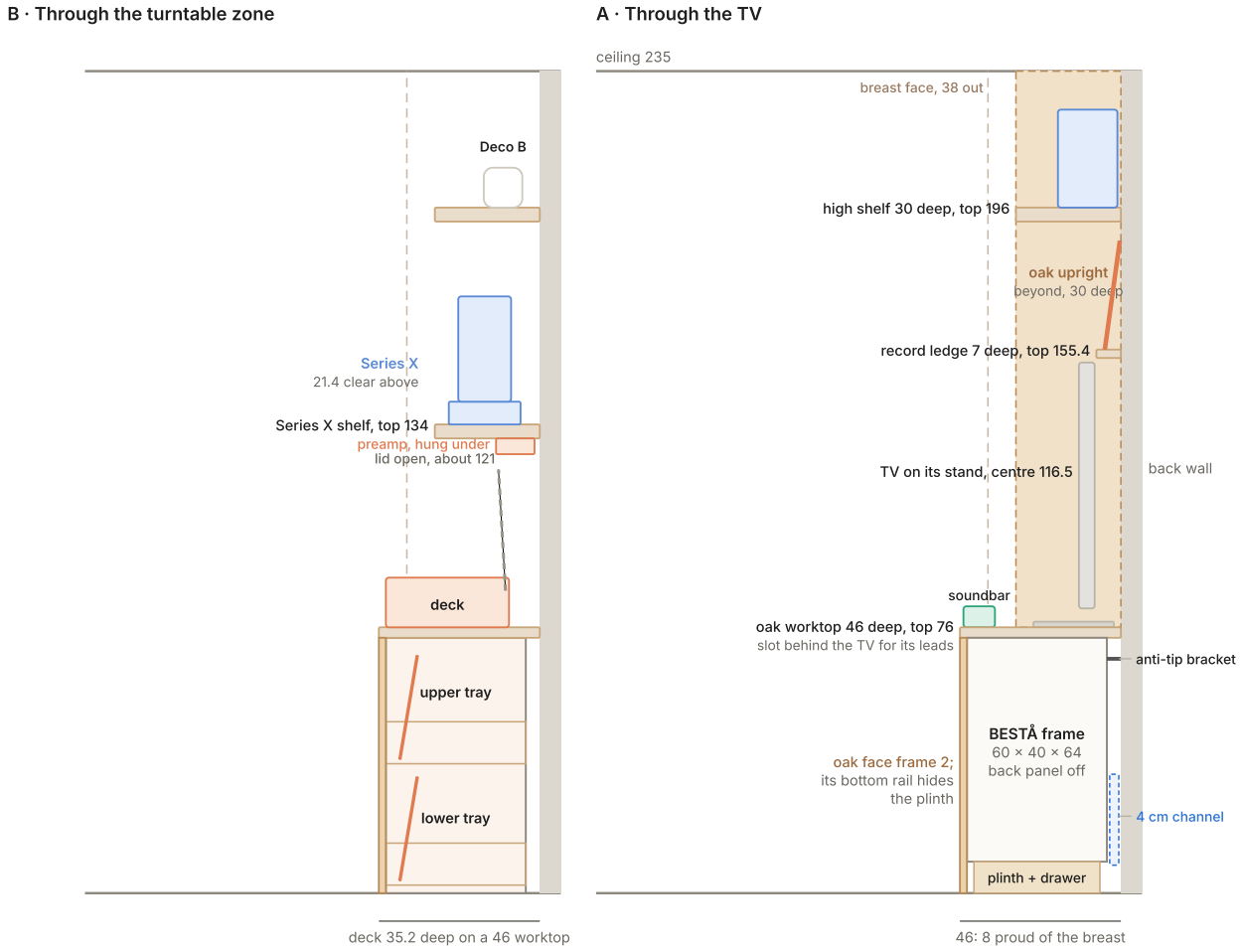
The media wall and the dining wall from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below; the fold-down table on the chimney breast. Heights in cm



How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



The sub in the media wall

You asked for the sub back in the unit, with the Xbox bricks moved behind. It fits, and it sits in the best spot for it: under the TV, beside the soundbar, so the bass comes from the same place as the picture. Here's how the base makes room, and what was checked.

- **The bricks go to the back.** The Xbox One's brick (17 × 7.5 × 4.8[11]) and the Xbox 360's (allowed 21 × 10 × 6[16]) lie flat at the back of their shelves, by the channel's power leads. Microsoft asks for a ventilated space, flat on the rubber pads, with nothing on top[26]; the bays are open front and back.
- **That frees bay 1's fronts** beside the two Xboxes, so the HDMI switch (beside the Xbox One, where its remote reaches it) and the RetroTINK (beside the Xbox 360) move up from bay 2.
- **Bay 2's lower half becomes two columns:** the sub on the left, under the TV, and the PS2 over the PS1 on the right. The sub is 21.4 × 30.4 × 29.4[3]; its column is 23.4 wide and 35.3 tall.
- **The port:** the sub's front sits flush with the face frame, so its rear port is 15.6 cm from the wall through the open back and the channel. That's inside the 15–30 cm advised for a rear port[25], at the low end, so the channel behind bay 2 stays clear: Switch 2 and the retro 6-way move behind bay 1.
- **Vibration, the one thing to test:** Microsoft advises keeping the Xbox 360 away from subwoofers[27], and the deck shares the worktop. The 360 is a whole bay away, the sub stands on an isolation riser without touching the divider or the frame, and the deck has isolation feet. Once it's built, play a disc game and a record at film volume. If anything skips, turn the sub down for that source; if it still does, the sub can go back under the dining wall.

- **The dining wall is clear underneath:** no cover to make, and nothing to move for eight at the table.

The fit proof: every item against its published size

Each zone below lists what goes in it, its size and where that size comes from (numbered sources, section 12), the space, what's left and a verdict. 49 of the 66 checks fit outright and none fail. 13 are tight: the deck's 2.65 cm sides and the preamp over the open lid; the sub's column, its port gap, the PS2 over the PS1 and the switch channel in bay 2; the dining seats at 55–57 cm each, the door's swing and the clearances for six and eight; and, if you choose the stairs wall for the table, its two end chairs. 4 need a check on site: the TV's plugs by the corner post, the RetroTINK's size when it arrives, whether the sub's vibration reaches the consoles and the deck (test it before fixing the shelves; the fallback is the dining wall), and, for the stairs option only, what the west wall is built of.

The TV between the uprights

126.5 between the oak uprights, the left one now a corner post on the side wall; the ledge's underside at 153

Item	Size (cm)	Space	Spare (cm)	Result
LG OLED C6 55 in, on its stand	122.2 × 70.3 × 4.5 panel; 75.7 tall on the stand [2]	126.5 wide	2.15 each side	**Fits**
Top of the screen	151.7 (bottom 81.4, centre 116.5) [2]	ledge at 153	1.3	**Fits**
LG stand base	47 × 23 [2]	worktop 46 deep	set 2 cm off the wall: 11 cm in front of it for the soundbar's leads	**Fits**
Soundbar, Hisense AX5100G, centred under the TV	92 × 9 × 6 tall [3]	x 21.75–113.75, top 82	the sightline to the screen's bottom passes 83.54–84.17 over it	**Fits**
The TV's HDMI plugs, with the screen 2.15 from the corner post	C6 sockets face sideways on the back panel [2]	behind the screen	plugs stay behind the panel; if a stiff lead pushes out, use 90° HDMI adapters	**Check**

The turntable zone, right of the TV

50.5 wide × 46 deep, worktop 76 to the Series X shelf's underside at 130

Item	Size (cm)	Space	Spare (cm)	Result
Turntable, AT-LP120X size (worst case)	45.2 × 35.2 × 14.2 with its lid [8]	50.5 wide	2.65 each side	**Fits, tight**
Its depth, with room behind for the lid's hinges	35.2, front 2 cm in from the worktop's edge [8]	46 deep, to the painted wall	8.8 behind it	**Fits**
Lid open	cover 44.1 × 34.6; standing up, its top at about 121 [17]	shelf underside 130	about 9	**Fits**
Phono preamp, hung under the shelf at the back	about 11 × 11 × 4.5	125.5–130, at the back	4.4 over the open lid's edge; lift the lid gently	**Fits, tight**
Series X fan stand on the shelf	20.5 × 20.5 (assumed; model to confirm)	50.5 × 30 shelf	15 each side	**Fits**
Xbox Series X, standing	15.1 × 15.1 × 30.1 tall [4]	top at 170.6	21.4 clear above (Microsoft's minimum is 4); 15 each side, inside the 10–15 it recommends	**Fits**
Deco B on the high shelf, right	11 × 11 × 11.4 [7]	50.5 wide, 39 to the ceiling	beside the framed 7-inch, 27 clear above	**Fits**

Shelves above the TV

ledge top 155.4, 7 deep; high shelf top 196, 30 deep; ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Four LPs on the record ledge	4 × 31.4 = 125.6 wide [13]	126.5 long	0.9: snug, as drawn	**Fits**
Ledge LPs leaning back 8°	tops at 186.5 [13]	high shelf underside 192	5.5 to lift one out over the 1.5 lip	**Fits**
Books on the high shelf	up to 30 tall	39 to the ceiling	9 over the tallest	**Fits**
Oak uprights	159 long (worktop to ceiling) [1]	235 ceiling	buy 165 and scribe to the ceiling	**Fits**

Bay 1, Xbox (BESTÅ frame on the 9 cm plinth)

56.8 wide inside, 40 deep; the bricks lie at the back of their rows

Item	Size (cm)	Space	Spare (cm)	Result
Xbox One, original, + HDMI switch	33.3 × 27.4 × 7.9; switch 12.7 × 5.9 × 2.5 [12,10]	16 tall row	8.1 above; the switch in the 23.5 beside it, at the front for its remote	**Fits**
Xbox 360, original, + RetroTINK-6X CE	30.9 × 25.8 × 8.3; RetroTINK allowed 23 × 23 × 8 (no size published) [12,9]	16 tall row	7.7 above; 25.9 beside it for the RetroTINK: measure it when it arrives	**Check**
Both bricks, flat at the back of their rows	Xbox One 17 × 7.5 × 4.8; Xbox 360 allowed 21 × 10 × 6 [11,16,26]	behind the switch and the RetroTINK, by the channel's power leads	23 + 10 = 33 of the 40 depth for the RetroTINK and the 360's brick; on their rubber pads, nothing on top, open to the channel; 2 cm or more from the consoles' rear vents	**Fits**
Game drawer	box 54.3 × 35 on 350 mm runners; cases 19 × 13.5 × 1.4 [15,13]	25.2 tall row, 40 deep	about 3 above the leaning cases; 5 behind the box	**Fits**
Depth with the leads	console 27.4 + HDMI plug about 5 [12]	40 deep, back open	about 7.6, then the 4 cm channel	**Fits**

Bay 2, PlayStation and the sub (BESTÅ frame on the plinth)

56.8 wide inside, 40 deep; under the PS3's shelf, the sub's column 23.4 wide, a 12 mm ply divider, then 32.2 for the PS2 over the PS1

Item	Size (cm)	Space	Spare (cm)	Result
PS3, original, + controller basket	32.5 × 27.4 × 9.8[12]	23.7 tall row, full width	13.9 above; 24.3 beside for the basket	**Fits**
The sub in its column	21.4 × 30.4 × 29.4 tall, on a 3 isolation riser[3]	23.4 wide × 35.3 tall	1 each side, 2.9 above; it mustn't touch the divider or the frame	**Fits, tight**
The sub's driver and port	front-firing driver, rear port; its front flush with the face frame[3,25]	15.6 from the port to the wall, through the open back and the channel	inside the 15–30 advised for a rear port, at its low end: keep the channel behind bay 2 clear (Switch 2 and the retro 6-way go behind bay 1)	**Fits, tight**
PS2, original, over the PS1	30.2 × 18.3 × 7.8 · 27 × 18.8 × 6[12]	32.2 wide; 16.5 and 17 rows on a part shelf	PS2 1 each side, 8.7 above; PS1 2.6 each side, 11 above; both open front and back	**Fits, tight**
Vibration: the consoles and the deck share the base with the sub	Microsoft: keep the Xbox 360 away from high-vibrating sources such as subwoofers[27]	the sub on its riser, clear of the divider and the frame; the deck on isolation feet	the 360 is a bay away, the PS2 and PS1 a divider away: play a disc game and a record at film volume; if anything skips, turn the sub down for that source, or put it back under the dining wall	**Check**
Hidden slim drawer in the plinth	box 52 × 34 × 5, 300 mm full-extension runners, a finger-pull groove (no push-to-open at floor level)[15]	9 cm plinth, 120 wide	about 2 above and below the box; felt its runners, it's under the sub	**Fits**
Switch 2 (TL-SG108) and the retro 6-way in the channel	15.8 × 10.1 × 2.5; 6-way about 34 × 6 × 4[6]	4 cm channel, behind bay 1	on their edges, clear of the sub's port	**Fits, tight**

Bay 3, the records (18 mm ply carcass from the floor)

56.4 wide × 40 deep × 69.4 tall inside, under the turntable

Item	Size (cm)	Space	Spare (cm)	Result
Two face-out record trays	trays 53.9 × 40 on 400 mm runners, 68–70 kg pairs[15,13]	56.4 inside	exact: 53.9 + 2 × 1.25 runner allowance = 56.4	**Fits**
Tray heights	lower sleeves to 34.4, upper to 69.1[13]	top panel underside 71.2	2.6 and 2.1 finger room	**Fits**
LP trough · singles trough	32.6 for 31.4 sleeves · 18.9 for 18.3–18.4[13]	52.1 inside each tray	0.6 each side · 0.25–0.3 each side	**Fits**

The side wall: the ledge wraps the corner, the PC shelf

one oak line at 155.4: ledge 134 long × 7, then a shelf 82 × 28

Item	Size (cm)	Space	Spare (cm)	Result
Four more LPs on the wrap ledge	4 × 31.4 + gaps = 129.8 [13]	134 long	2.7	**Fits**
The PC on its shelf	23 × 45.3 × 46.6 tall (an average gaming case, e.g. Corsair 4000D) [29]	28 × 82	2.5 off the wall, the brush grommet and pad dock beyond it; height 46.6 under the 79.6 to the ceiling	**Fits**
Air above the PC	top at 202	235 ceiling	33 for the top exhaust	**Fits**
Shelf over the end of the sofa	underside 151.4 [24]	sofa back 80	71.4; the shelf covers only the 28 cm nearest the wall	**Fits**
Keyboard tray under the PC shelf (v6.1)	7 cm housing, tray rolls 25 out on 300 mm runners [15]	sofa back 80	64.4 over the sofa's back; pulled out, its front edge is 53 from the wall, still over the return's seat	**Fits**
Out of the TV's sightlines	PC about 54° left of the line to the screen from the sofa [1]	—	v10's PC sat at the screen's left edge	**Fits**

The dining wall on the chimney breast

breast 117.5 wide; the table 110 × 80 × 2.7 solid oak, about 17 kg

Item	Size (cm)	Space	Spare (cm)	Result
Table on the breast	110 wide [1]	117.5	3.75 each side	**Fits**
Folded up	about 6 off the breast, top edge 153.3 [19]	framed LPs from 165	11.7	**Fits**
Two Hebgo brackets under it	arms 58, 80 apart; 500 kg a pair [18]	a 80 deep top	22 beyond the arms; spacing inside Häfele's 80	**Fits**
Three lift-off hinges on the backboard	100 × 88, rated for a 120 kg door [19]	a 17 kg top	slide it 5 cm sideways to lift it off	**Fits**
Four at the wall: two on the room side	110 of edge [23]	60 each recommended	55 each	**Fits, tight**
Four at the wall: room to the sofa	table edge 83 out [23,24]	sofa front	121 (walk behind: 100+)	**Fits**
The door and the dining chairs	door 83 wide, hinged on the breast side, opens in (you confirmed) [1]	its swing reaches about 44 out from the recess wall	the west-end chair (four) and the breast-side chairs (six, eight) sit in its swing: keep it shut while you eat, or have it rehung to open into the hall	**Fits, tight**
Six and eight: lifted off onto legs, with the leaf	table 110 + leaf 60 = 170 × 80, top at 73.7 on the oak legs [20]	dining height about 75	within 1.3; felt glides take up an uneven floor; four legs under the table, four under the leaf, joined by two dowels and two clips	**Fits**
The eight legs, stored in the leaf's frame (v13.6)	8 legs 4.5 square in clips 5.6 wide at 6 centres = 47.6 across; each 71 + 1.5 bolt long [21,20]	inside the frame 50.4 × 73.4, 5.5 deep	1.4 each side, 0.9 along, 1 below the legs; leaf, frame and legs about 16 kg together	**Fits**
Knees under the leaf's frame	frame underside at 65.5 with the table on its legs [22]	chair seat 48	17.5 above the seat, like an ordinary table's apron; the 110 top has no frame (23)	**Fits**
Six: two a side, one at each end	170 of edge each long side, 80 at each end [23]	60 each	85 each on the sides (25 spare); 80 at the ends	**Fits**
Eight: three a side, one at each end	170 long [23,20]	60 each	56.67 each	**Fits, tight**
Six and eight: to the breast and the window	table 75–155 out [23]	75 to aim for	75 to the breast; 154 to the window	**Fits**
Six and eight: the west end	table end at 402 [23]	west wall at 467	65 for the end diner (60 is the tight minimum; the stairs are behind the breast-side corner, clear of it)	**Fits, tight**
Six and eight: the east end and the coffee table	end chair at x 183–232 [23,24]	coffee table and sofa	7.5 to the coffee table's edge: slide it towards the window first; the sofa ends at 190, clear of the chair (49 to its front line)	**Fits, tight**
Chairs at the table	seat 48 [22]	top 76	28 seat to top, the usual 25–30	**Fits**

Feet under the table, four at the wall	nothing under it: the sub is in the alcove base [22]	table 80 deep	80 of clear legroom along the breast; eight need no cover lifted off	**Fits**

Option: the table under the shelf, by the stairs

the same 110 × 80 fold-down table on the west wall, centred 180 from the breast's face line

Item	Size (cm)	Space	Spare (cm)	Result
The wall beside the stairs	table 110 long [28,1]	268 from the stair box to the window wall	84 to the stair box, 74 to the window wall	**Fits**
Folded, under the shelf	top edge 153.3 [28]	shelf's underside about 195	41.7: the shelf and what's on it stay	**Fits**
Down for four: the room side	table edge 83 out from the wall; chairs to x 335 [23,22]	the sofa's end at x 190	145 to walk behind the chairs (100+ is the aim)	**Fits**
The stair end's chair	49 deep at the table's south end [22]	84 to the stair box	35 behind it: tuck it in after a meal so the stair's foot stays clear	**Fits, tight**
The window end's chair	49 deep at the north end [22]	74 to the window wall	25 behind it	**Fits, tight**
The door's swing	the door opens in, 83 wide [1]	its edge reaches -44	81 clear of the table: unlike the breast, nobody sits in its swing	**Fits**
Light over the table	the existing pendant point, for bar 2 [28]	point about (412, -143)	right over the table's stair end: the bar's heads light the table directly	**Fits**
Fixing the brackets	500 kg a pair, into the wall [18]	the west wall's build isn't known yet	solid (plugs and screws) or studwork (fix into studs): knock and look before choosing this site	**Check**

The room

467 × 347, ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Eyes to screen from the sofa	the L-sofa's long side against the window wall, 10 off the radiator [1,24]	335 window wall to screen	about 2.65 m; a 55-inch fills about 26°	**Fits**
From the end of the sofa's return	sitting up at its end, 217 from the screen [24]	straight ahead of the TV	the TV's move puts it square on to this seat	**Fits**
TV centre against seated eyes	116.5 [2]	eyes at 105–112	about 1–2.5° above the eye line	**Fits**
Both spotlight bars under the ceiling	22.4 drop [14]	235 ceiling	lowest shade at 212.6	**Fits**

Console sizes, every version

The bays are sized for the largest version of each console. The model number is on the label underneath; a smaller version just leaves more room.

Console	Version	W × D × H (cm)	Power

PS1	Original (largest) · PS one	27.0 × 18.8 × 6.0 · 19.3 × 14.4 × 3.8	Built in · external adapter
PS2	Original (largest) · Slim	30.2 × 18.3 × 7.8 · 23.1 × 15.2 × 2.8	Built in · external brick
PS3	Original (largest) · Slim · Super Slim	32.5 × 27.4 × 9.8 · 29.0 × 29.0 × 6.5 · 29.0 × 23.0 × 6.0	Built in
Xbox 360	Original (largest) · S · E	30.9 × 25.8 × 8.3 · about 27.0 × 26.4 × 7.5 · 25.4 × 26.4 × 6.7	External brick
Xbox One	Original (largest) · S · X	33.3 × 27.4 × 7.9 · 29.5 × 22.9 × 6.4 · 30.0 × 24.0 × 6.0	External brick · built in
Xbox Series X	Standing	15.1 × 15.1 footprint, 30.1 tall	Built in

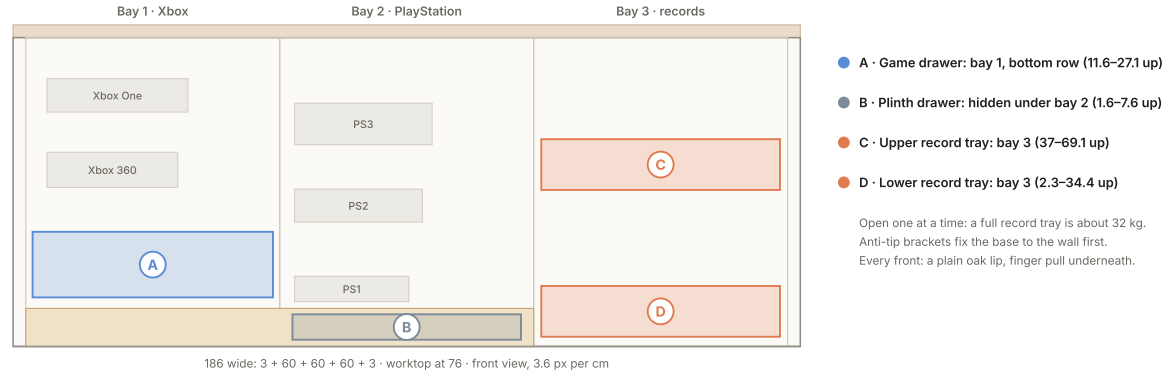
The drawer layouts

Four drawers in the alcove base, all drawn to scale from above: **A** the game drawer (bay 1), **B** the hidden plinth drawer under bay 2, **C** and **D** the two face-out record trays (bay 3). Each one shows its compartments, what goes in them and how many, and its runners. Every front is a plain oak lip with a finger pull cut underneath.

The drawer layouts: where each drawer is and what goes in it

Four drawers in the alcove base, all to scale: A the game drawer, B the hidden plinth drawer, C and D the two face-out record trays

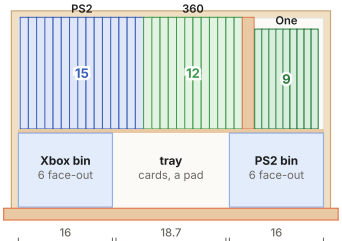
Where each drawer is: the alcove base from the front



186 wide: 3 + 60 + 60 + 60 + 3 · worktop at 76 · front view, 3.6 px per cm

A · Game drawer

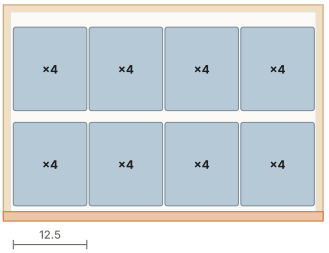
box 54.3 × 35 · 350 mm soft-close runners · rolls out 30



back: 36 spine-up, titles on top (19 cm cases on edge)
front: 12 face-out · about 48 games in all

B · Hidden plinth drawer

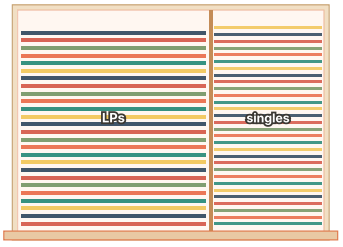
box 52 × 34 × 5 inside · 300 mm runners · finger pull



32 PS1 games flat: 8 stacks of 4 jewel cases
or remotes, spare pads, cables and manuals

C and D · Record trays (the two are the same)

box 53.9 × 40 · 400 mm runners, 68 kg a pair



every sleeve face-out, leaning back 10°: flip like a shop
~90 LPs + ~120 singles each · two trays

Plans drawn from above at 5 px per cm, front (the room) at the bottom; the elevation at 3.6 px per cm. Sizes: BESTÅ frame 60 × 40 × 64 (IKEA);
game cases 19 × 13.5 × 1.4 and 17 × 13.5 × 1.2; jewel case 14.2 × 12.5 × 1; sleeves 31.4 and 18.3; runners Emuca 400 mm 68 kg, Accuride 350 mm (see the fit proof).



Every drawer out, in place: the game drawer and the upper record tray 30 cm out, the plinth drawer 25 cm out. In use, open one record tray at a time (a full one is about 32 kg).

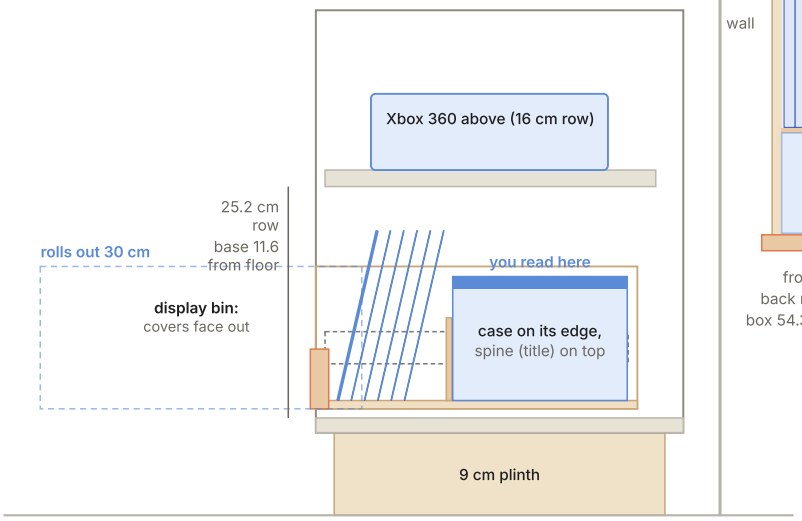
The game drawer (bay 1, bottom row)

Unchanged. Two display bins at the front show covers face-out even with the drawer shut (Xbox left, PS2 right, a small tray between them); behind them the rest of the games stand on edge with the spine on top, so every title faces up when you roll it out.

The game drawer: covers on show at the front, every title readable from above at the back

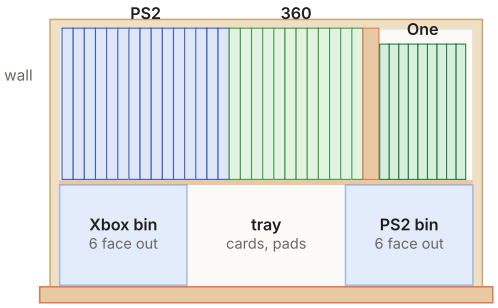
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out



front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

- **Box:** about 54.3 × 35 cm outside on 350 mm full-extension soft-close runners screwed to the BESTÅ sides; sides 15 cm; 9 mm base lined in felt. About 48 games.
- **Runners:** Accuride DZ3832EC (43 kg) or Screwfix Essentials 964VN (40 kg); not IKEA's BESTÅ runners, which aren't full extension.
- **The bricks:** both Xbox bricks lie flat at the back of their shelves, on their rubber pads, beside the channel's power leads, with nothing on top. Microsoft only asks that the space is ventilated[26], and these bays are open at the front and the back.

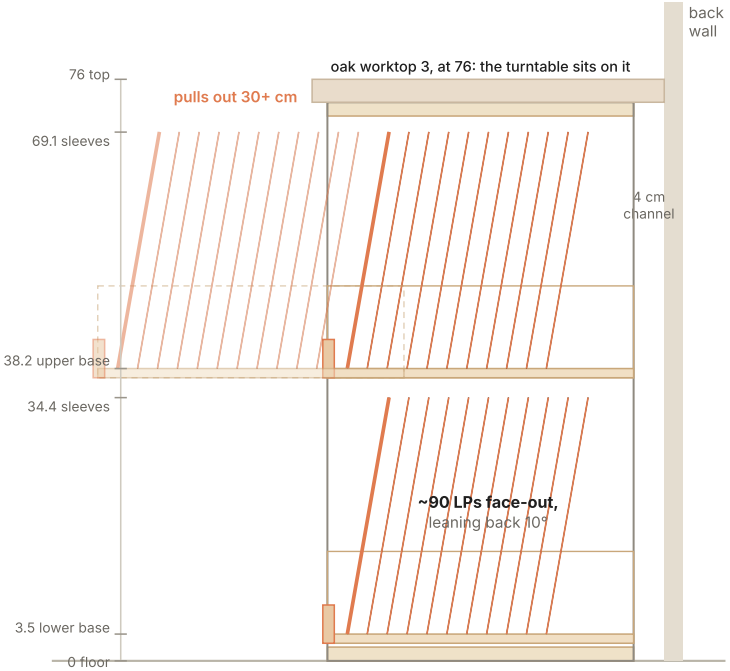
4 · The records and the turntable

The vinyl is back in one place, in the alcove: the deck on the worktop, the records in the two trays right underneath it, the preamp tucked under the shelf above. You can pull a tray, pick a record and play it without crossing the room.

The record trays: every LP and single face-out, two tiers in bay 3

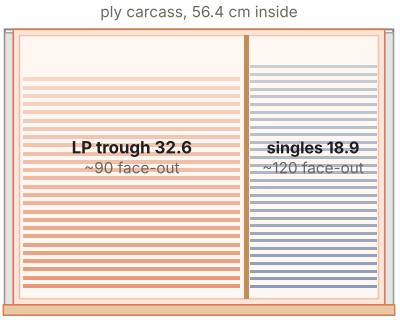
Bay 3 of the alcove base, right under the turntable. Section left, plan right. Pull a tray and flip through the covers like a record shop browser

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · trays 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Top view of one tray



tray 53.9 × 40 outside, 52.1 × 38.2 inside
9 mm sides · 6 mm oak divider · 5 cm oak lip
about 32 kg full: open one tray at a time

How the record trays work

- **LP trough:** 32.6 cm wide, so a bare 31.4 cm sleeve has 6 mm each side. Records lean back 10° like a record shop browser; the last one leans on a raked oak rest.
- **Singles trough:** 18.9 cm wide, the same lean.
- **Capacity:** about 90 LPs and 120 singles per tray; two trays, about 180 LPs and 240 singles.
- **Runners:** 400 mm full extension, 68–70 kg per pair, screwed straight into bay 3's 18 mm ply sides. A full tray weighs about 32 kg: open one at a time, and fit the anti-tip brackets before loading.
- **Outer sleeves:** if most LPs live in thick outer sleeves (up to 32.3 cm), widen the LP trough to 33.4 cm; the singles trough then drops to 18.1, too narrow for face-out singles.

Dimension	Figure	How it adds up
Bay 3 carcass	60 × 40 × 73 outside, 56.4 × 69.4 inside	18 mm birch ply; back open to the channel
Tray	53.9 × 40 outside, 52.1 × 38.2 inside	56.4 − 2 × 1.25 runners; 9 mm sides and ends
Inside layout	32.6 + 0.6 + 18.9	LP trough, oak divider, singles trough
Lower tray · upper tray	Undersides 2.3 · 37.0; sleeves to 34.4 · 69.1	2.6 and 2.1 cm finger room; top panel at 71.2

The turntable zone

- **The deck:** on isolation feet, 2 cm back from the worktop's front edge, with 8.8 cm behind it for the lid's hinges. The zone takes a deck up to the size of an AT-LP120X (45.2 × 35.2 cm); most decks are smaller.
- **The lid:** AT's cover is 44.1 × 34.6 cm[17], so standing open its top edge is at about 121; the shelf's underside is at 130.
- **Bass:** the sub is in bay 2, in the same base the deck sits on. It stands on an isolation riser, clear of the divider and the frame, and the deck stands on isolation feet. If a record skips or hums at film volume, turn the sub down for vinyl.
- **Signal path:** turntable set to PHONO → the preamp under the shelf → a 2 m RCA-to-3.5 mm lead down the groove behind the right upright and along the back of the worktop to the soundbar's AUX. If the deck has no PHONO/LINE switch, plug it straight into the AUX.
- **Soundbar:** Music mode, bass up, surround, night and dialogue modes off. Always the AUX cable, never Bluetooth.

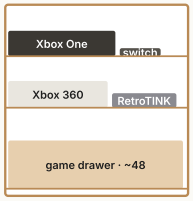
Storage options, to scale

Storage options: what fits each space, to scale

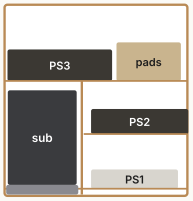
Front views and views from above of every storage space in the wall, with counts, and where the overflow would go

Bays 1 and 2: inside each BESTÅ frame

56.8 wide × 40 deep inside; rows as planned, front view



Bay 1 · Xbox



Bay 2 · PlayStation + sub

Swap any 16 cm row for two KUGGIS boxes, 26 × 35 × 15, side by side (52 of 56.8 wide).

Bay 3: each record tray, from above

53.9 × 40 outside, two trays; everything face-out



LP trough 32.6 · singles 18.9

≈ 90 LPs + 120 singles a tray

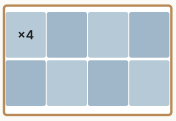
× 2 trays = 180 LPs, 240 singles

sleeves lean back 10°, flip like a shop

thick outer sleeves: widen the LP trough to 33.4; singles drop to 18.1

Hidden plinth drawer, from above

box 52 × 34 × 5 inside, under bay 2; finger pull, not push-to-open



32 PS1 games flat

jewel cases 14.2 × 12.5 × 1, 8 stacks of 4

or: remotes, spare pads, cables, manuals

Series X shelf, front view

50.5 wide, 134 to the high shelf at 192: 58 clear



15 air each side, 21.4 above

Microsoft: at least 4 above, 10–15 recommended all round

Under it: the deck, lid opens to 121

High shelf, front view: as planned, or as LP overflow

186 long, 30 deep, 39 clear to the ceiling; the right-hand 50.5 keeps Deco B and the framed 7-inch



about 1 m of books, a lidded box, a print

As planned



≈ 380 LPs spine-out at about 3.3 mm each; a 31.4 sleeve overhangs the 30 cm shelf by 1.4

As LP overflow

PC shelf, from above

82 along the wall × 28 out, at 155.4; keyboard tray in a 7 cm housing under it



side wall along the top · alcove to the right

Keyboard tray: 300 mm runners, rolls 25 cm out; the housing stops 12 short of the sofa end so the cables drop past it. Full-size wireless keyboard and mouse.

If the collection outgrows the wall

two overflow options, both face-out or spine-out to match

1 · LPs spine-out on the high shelf: about 380 more

No building; books move to the second option's top.

2 · A second record unit, west of the sofa, under the window

Bay 3 again (60 × 46 × 73, two face-out trays): 180 LPs, 240 singles. At x 300–360 it leaves 123 cm to the table set for six (100 to walk).



front view, 60 × 73 (shown at half height)

All to one scale, 2.2 px per cm. Sizes: KUGGIS box 26 × 35 × 15 (IKEA); jewel case 142 × 125 × 10 mm (Wikipedia: Optical disc packaging); consoles, sleeves, Series X and the trays from the fit proof (section 3 of the plan).

Where every record lives

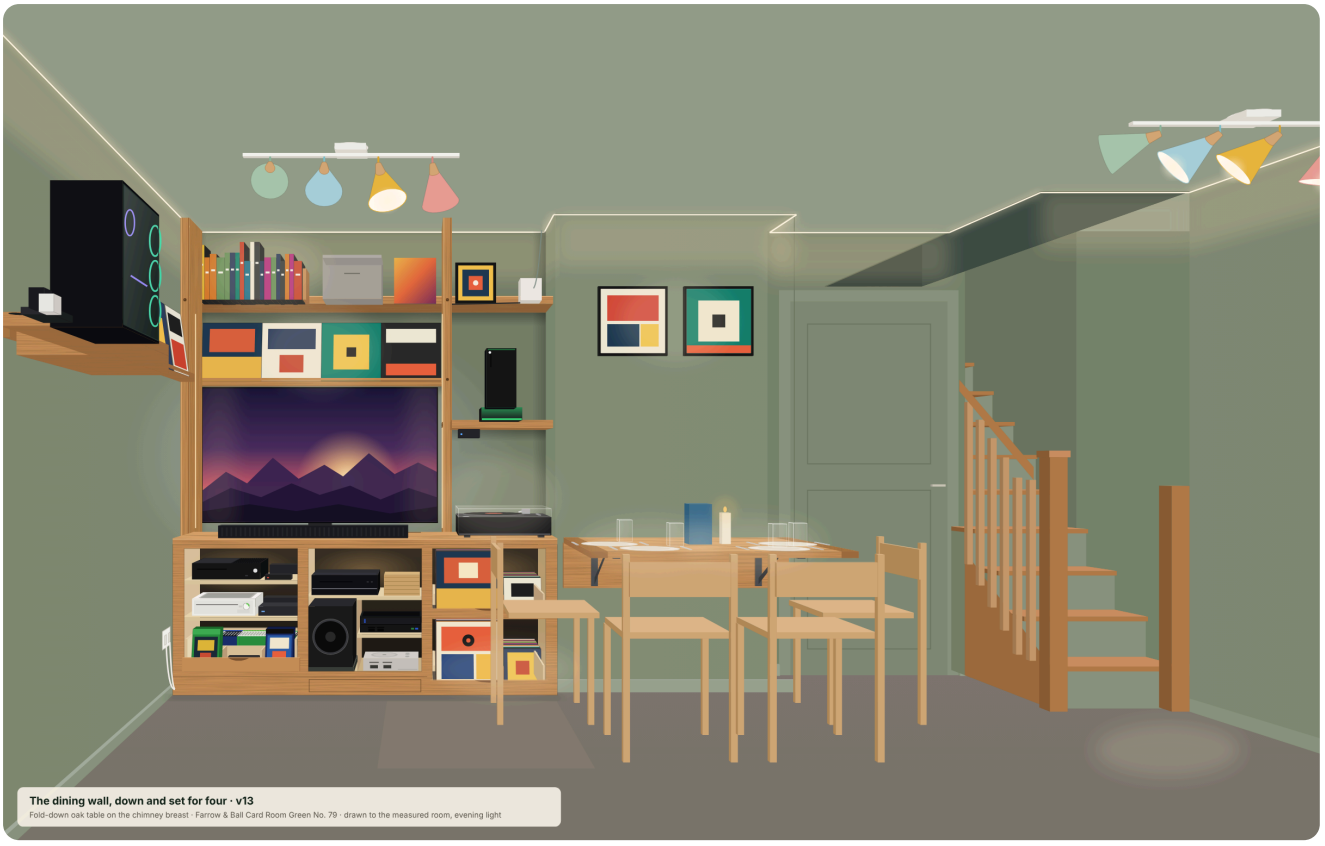
Where	What	How many
Bay 3, upper and lower trays	LPs and singles, face-out	About 180 LPs + 240 singles
Above the TV	Oak record ledge, the "now playing" wall	4 LPs

Side wall	The ledge wrapped round the corner	4 LPs
High shelf, right	A framed 7-inch	1
Total		**About 188 LPs and 240 singles, every one face-out**

That's fewer than v10's cabinet held (it had a 50-LP browser and a singles drawer). If the collection is bigger than this, tell me the counts. The neatest overflow is a matching face-out tray unit under the window at the end of the sofa.

5 · The dining wall

The chimney breast becomes the dining wall. A solid oak table, 110 × 80 cm, hangs on lift-off hinges on an oak backboard. Every day it stands folded flat against the breast, about 6 cm deep. Down, it's a table for four at the same 76 cm as the alcove's worktop. Lifted off with its matching 60 cm leaf, which carries all eight oak legs in a slim frame underneath, it makes a 170 × 80 table in the room: six with room to spare, or eight at a squeeze.



Down and set for four: two on the room side, one at each end, nothing under the wall now. The second spot bar, on the point by the door just out of view here, lights the table and the framed LPs.

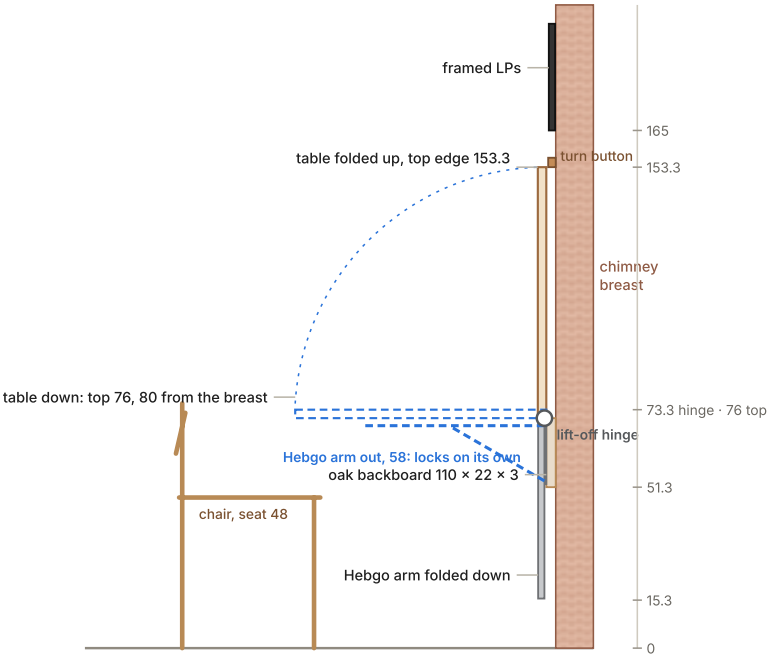


Lifted off with its leaf (170 × 80, on eight tapered oak legs; the leaf's slim frame shows under its end) and set for six, 75 cm off the breast: two a side with 85 cm each, one at each end. The camera stands back in line with the alcove's edge, so both spot bars show: bar 1 on the rose over the coffee table, bar 2 on the point by the door, two heads on the table. The backboard and the folded brackets stay on the breast. The stairs (going up) and the door are drawn roughly until you confirm the west end.

The dining wall in section: folded, down, or off the wall

Through the chimney breast at the table's centre, to scale, room on the left. Solid: folded. Dashed blue: down for four

- Everyday: folded flat, about 7 cm off the breast, held at the top by two turn buttons.
- For four: lift the top edge a little, turn the buttons, lower it onto the two brackets as they swing out and lock.
- For six or eight: with it down, slide it 5 cm sideways off the lift-off hinges' pins, screw in four oak legs from the leaf's frame,
- and stand it in the room with the leaf: the top is then 73.7 cm. Two people lift it: about 18 kg.



Two places for the table: the chimney breast, or under your shelf by the stairs (v14)

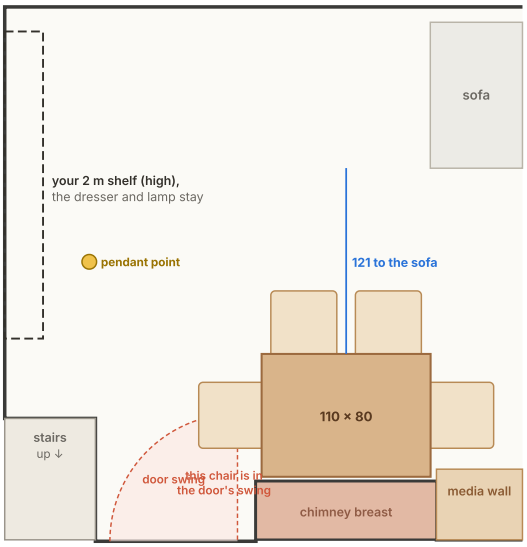
You asked what it would look like with the table on the wall going up the stairs, under your 2 m shelf, with everything else free to move and the floor lamp staying but able to shift. Your photos put that wall at about 268 cm from the stair box to the window corner, with the shelf about 200 cm up. The same table, backboard, brackets and leaf work on either wall: only the backboard's fixings change. Option B is drawn in the 3D room on the plan page, from every angle and in each lighting scene.

Where the dining table can live: the chimney breast or the stairs wall

Both to scale, set for four at the wall, the west half of the room from above (window at the top). Six and eight lift it off onto legs in the same spot either way.

A · On the chimney breast

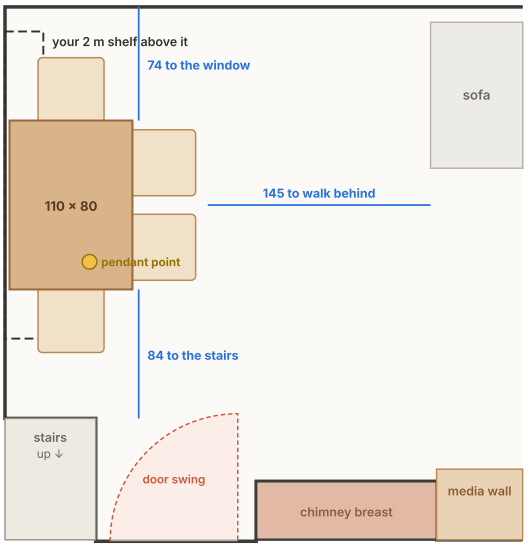
the plan so far: down at 76 for four, folded flat every day



+ in line with the worktop at 76; next to the media wall
+ the stairs wall keeps your shelf, dresser and lamp
– the west end's chair sits in the door's swing
– the breast's framed LPs sit above a busy wall

B · Under your shelf, by the stairs

the same table and brackets on the stairs wall; the dresser moves out



+ right under the pendant point; out of the door's swing
+ frees the chimney breast (frames, the lamp or a bench)
– the end chairs are tight: 35 to the stairs' foot, 25 to the window
– the wall's build is unknown: check it before fixing 500 kg brackets



Option B in the 3D room, set for dinner: bar 2 hangs on the existing pendant point over the table's stair end, your shelf stays above it, the stairs are behind the end chair.



Option B folded on an everyday evening: about 6 cm deep against the wall, 42 cm under the shelf, with the floor lamp beside it and the chimney breast left clear.

- **B fits:** centred on the wall it leaves 84 cm to the stair box and 74 cm to the window. Folded, its top edge (153.3) is 41.7 cm under the shelf, so the shelf and what's on it stay.
- **B's wins:** nobody sits in the door's swing; the existing pendant point is right over the table, so spot bar 2 lights it directly; the chimney breast is left clear (framed LPs, the lamp, or an armchair); and with the dresser gone the room's west half opens up.
- **B's costs:** the two end chairs are tight (35 cm behind the stair end's, 25 behind the window end's), so tuck them in after a meal to keep the stair's foot clear; six and eight still happen in the middle of the

room, as on the breast. The west wall's build is unknown: knock on it before choosing (solid takes plugs; studwork means fixing the backboard into the studs).

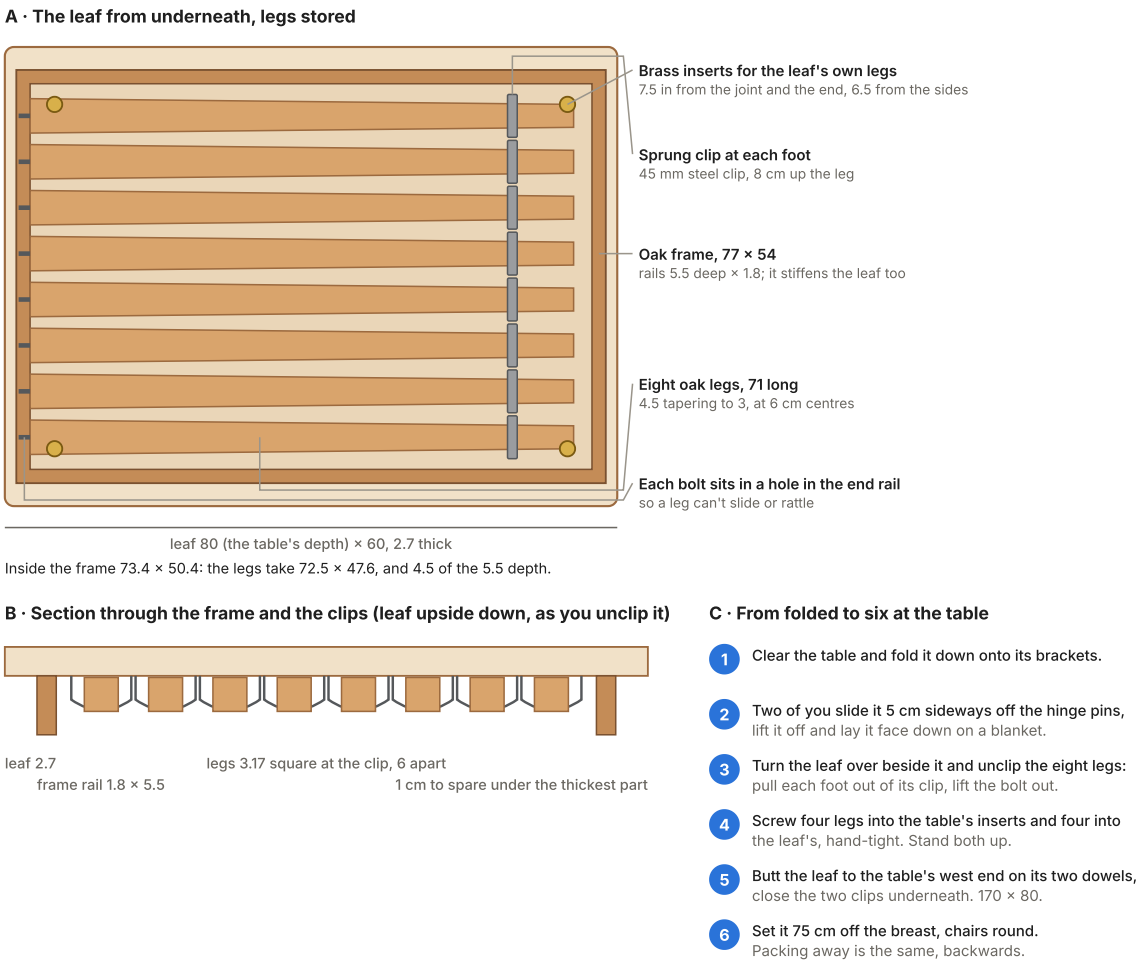
- **What moves for B:** the dresser goes (its turntable and records already have homes in the media wall: face-out and spine-up); the floor lamp moves to one of the spots in the 3D room's options; the sofa stays where it is, with a clear walk behind the chairs.

Where the legs live: inside the leaf (v13.6)

You asked for the legs to be part of the table's build, maybe in a box, so they don't become one more thing to store. The leaf is the box. A slim oak frame, 77 × 54 with rails 5.5 deep, is glued and screwed under the 60 × 80 leaf. All eight legs lie side by side inside it: each one's bolt sits in a hole in the end rail and its foot snaps into a sprung clip[21]. Leaf, frame and legs come to about 16 kg and go wherever the chairs are kept.

The leaf carries the legs: a slim oak frame underneath holds all eight

To scale. A from underneath, B in section, C the steps from folded to six



Why the leaf: the folded table stays a plain oak panel, nothing new stands in the room, and the leaf, with every leg inside it, goes wherever the chairs are kept. Leaf, frame and legs together weigh about 16 kg.
Set aside: folding legs under the table (they'd show on the folded panel and stand it about 5 cm further off the wall), a box under the dining wall (where the breast-side chairs push back with six seated), and the plinth drawer (too short for 71 cm legs).

- **Why this way:** the folded table stays a plain oak panel, nothing new stands in the room, and the frame stiffens the leaf like an apron. With the table up for six, the frame's underside is 65.5 cm from the floor, 17.5 above a chair seat: the same as an ordinary table's apron.
- **What we set aside:** folding legs fixed under the table (they'd show on the folded panel and stand it about 5 cm further off the wall); a box under the dining wall (it sits where the breast-side chairs push back with six seated); the plinth drawer (too short for 71 cm legs).

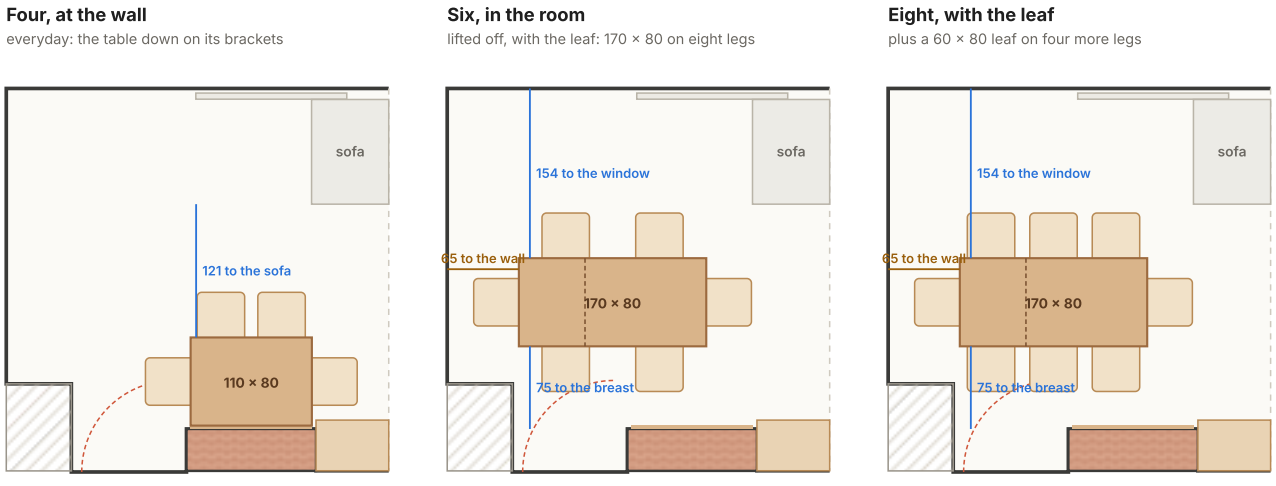
- **The legs are oak now,** not the steel option: they match the wall, and the case is sized to them (4.5 tapering to 3, 71 long).

How it works

- **The top:** solid oak with **pegged breadboard ends**, 7 cm oak strips across each end on a tongue, pinned with three oak pegs in slotted holes so the top can move. They stop it cupping when it lives upright, and they frame it like the uprights frame the TV. The corners are rounded to a 5 cm radius (kinder on hips by a door), and the underside edges are chamfered 1 cm so it reads slim.
- **Backboard:** oak, 110 × 22 × 3, from the worktop's offcut, screwed to the breast with plugs every 20 cm, top at 73.3.
- **Brackets:** two Häfele Hebgo heavy-duty folding brackets (287.43.428: 580 mm arm, 500 kg a pair[18]), 80 cm apart on the backboard's face. Raised, they lock on their own; a light lift on the arm releases them, and they fold flat down the breast.
- **Hinges:** three 100 mm lift-off hinges[19], all the same hand, along the backboard's top edge. Slide the table 5 cm sideways and it comes off.
- **Folded:** two oak turn buttons on the breast, plus a magnetic catch, hold the top edge at 153.3. The underside faces the room: plain oiled oak between the pegged breadboard ends, the same finish as the worktop, so folded up it reads as a calm oak panel under the two framed LPs (v13: no reeding). The legs screw into four small brass threaded inserts in the breadboard ends instead of black plates.
- **For six or eight:** lift it off (two people, about 18 kg) and lay it face down on a blanket. Unclip the eight tapered oak legs from the leaf (4.5 cm to 3 cm, 71 long, the standard dining leg height[20]), screw four into the table's brass inserts and four into the leaf's, and stand both up: the top is at 73.7. Butt the 60 × 80 leaf against the table's west end, joined by two alignment dowels and two table clips: six and eight both use it (v13.1: 110 × 80 alone gives six only 55 cm each, under the 60 a diner needs). Felt glides take up an uneven floor.
- **Chairs:** drawn at IKEA TERJE size (43 × 49, seat 48[22]), which folds flat. Any dining chair with a 45–48 cm seat works.

The dining table for four, six or eight

To scale, the west half of the room from above, window at the top. Red: the door's swing. Blue: the clearances that matter



- Per diner: 60 cm of table edge (55 at a squeeze). Table edge to a wall: 60 cm is a tight minimum, 75 to aim for, 100+ to walk behind someone seated.
- Six and eight: slide the coffee table to the window first. The dashed red arc is the door's swing if it opens into the room: the chairs on that side need it shut while you eat.
- Chairs drawn at IKEA TERJE size (43 × 49, seat 48): 28 cm from seat to table top, the usual 25–30.

Seats	Where the table stands	Per diner	Clearances
Four	Down at the wall, on its brackets	55 on the room side (60 is ideal); the ends are 80 wide	121 to the sofa: room to walk behind
Six	Lifted off with the leaf: 170 × 80 at x 232–402, 75 off the breast	85, two a side; 80 at each end	75 to the breast (the figure to aim for), 154 to the window, 65 to the west wall
Eight	The same 170 × 80, in the same place	57, three a side	As for six: nothing to move, the wall is clear underneath

Clearances from a UK planning guide: 60 cm of table edge per diner, 60 cm from the table to a wall as a tight minimum, 75 to aim for, 100–110 to walk behind someone seated[23]. For six or eight, slide the coffee table to the window first, and keep the door to the hall shut while you eat if it opens into the room: the chairs on that side sit in its swing.

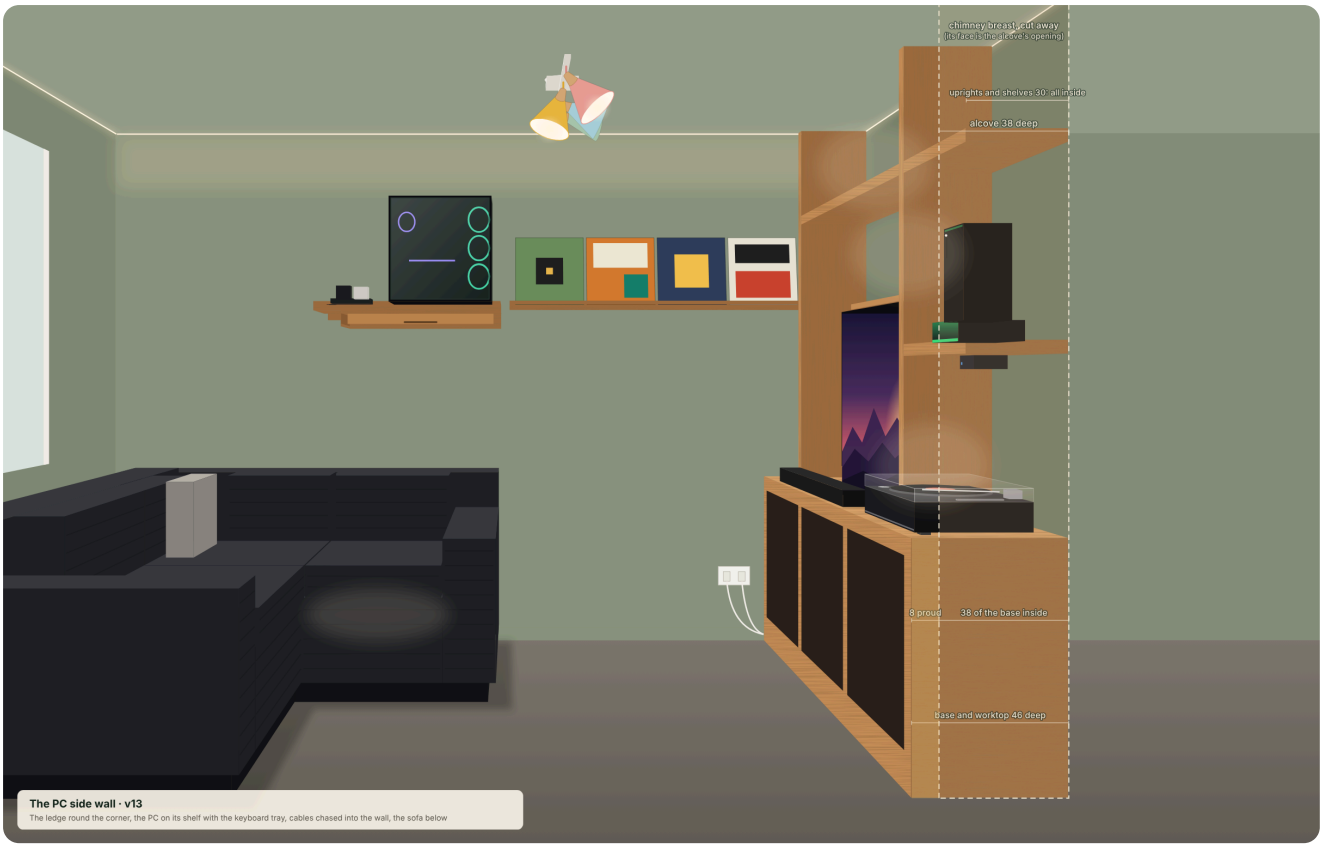
Cut list and hardware

Part	Material	Size (cm)	Qty
Table top · leaf	Solid oak, edge-glued staves	96 × 80 × 2.7 centre + two 7 × 80 breadboard ends (110 overall) · 60 × 80 × 2.7	1 · 1
Breadboard pegs	10 mm oak dowel, slotted holes in the tongue	3 per end	6
Legs	Solid oak, tapered on two inside faces	71 × 4.5 tapering to 3, an M8 hanger bolt in each	8
Leaf frame (the leg case)	Solid oak	long rails 77 × 5.5 × 1.8; side rails 50.4 × 5.5 × 1.8, one with eight 10 mm holes	2 + 2
Leg clips	RS PRO 45 mm steel spring clips	42.1 × 56 × 12.7 mm	8
Backboard	Solid oak, from the worktop offcut	110 × 22 × 3	1
Turn buttons	Oak offcut	6 × 2.5 × 2.2, on a single screw	2
Leg inserts	Brass M8 threaded inserts, in the breadboard ends and the leaf	—	8
Folding brackets	Häfele Hebgo 287.43.428	arm 58, height 22	2
Lift-off hinges	Union PowerLoad, 100 × 88	one 3-pack, same hand	3
Leaf joint	Alignment dowels, table clips	—	2 + 2

Safety: never fold it up with anything on it, keep both turn buttons turned when it's up, and check the brackets have locked (a click each side) before you load it.

6 · The side wall

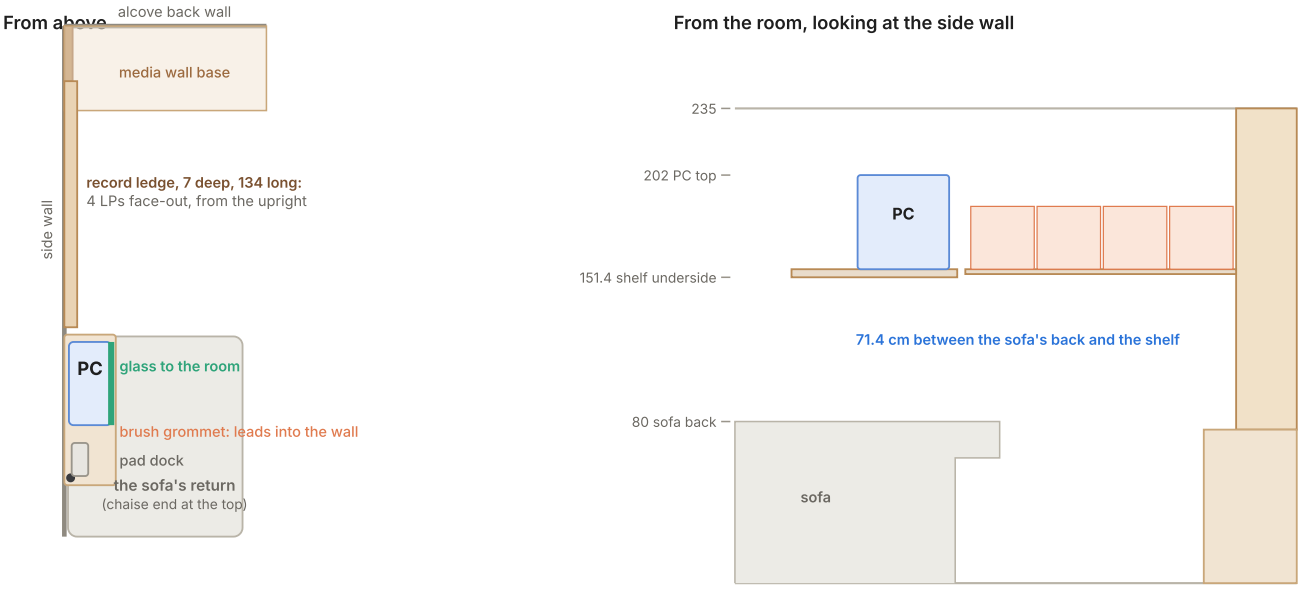
One oak line at 155.4 cm runs from the ledge over the TV, round the corner on the left upright, and along the side wall: first a 7 cm record ledge with four more LPs face-out, then an oak shelf for the PC over the end of the sofa. It ties the alcove to the rest of the room and moves the PC out of the TV's sightlines.



The side wall from the room: the ledge round the corner with its four LPs, the PC on its shelf with the keyboard tray's housing underneath, and nothing else on the wall: the leads run inside it, the sofa below (drawn from your photo), and the media wall end-on at the right. The chimney breast is cut away so the wall reads end-on; its outline is dashed, with the depths marked: the alcove is 38 deep, so 38 of the 46 cm base sits inside it and 8 stands proud (picked out in gold), and the 30 cm uprights and shelves sit wholly inside.

The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines



- **Wrap ledge:** 134 × 7 × 2.4 cm with a 1.5 cm lip, on an oak cleat screwed to the side wall, butt-jointed to the left upright's front edge on two 10 mm dowels.
- **PC shelf:** 82 × 28 × 4 cm, on an oak cleat with two concealed heavy-duty brackets (40 kg+ a pair), 71.4 cm above the sofa's back. The case sits with its glass to the room and its front towards the alcove. The ports face the window and stay open to the air (v13: no screen, so nothing traps the exhaust); from the sofa you look at the glass side, not the ports. The pad dock sits beyond the case.
- **Cables:** through a brush grommet at the shelf's back corner straight into a 25 mm conduit chased into the side wall (it's solid, as you confirmed), down to skirting height and behind the skirting to the base's plinth; you chase it yourselves in step 3, with a second, smaller conduit beside it for the PC's mains lead, then it's filled and painted, so nothing shows. A draw wire left in it pulls the next lead through. The HDMI is an active optical cable (about 5 m), "source" end at the PC.
- **Power:** the PC plugs into the 10-gang behind the TV: its lead (a 5 m C13 lead with a rewireable plug, fed down bare and the plug fitted at the bottom) runs in its own 25 mm conduit beside the data one. The 10-gang's load check, with the PC, is in section 9. No new socket.
- **Keyboard tray (back from v6.1):** the shelf sits on a 7 cm oak housing with a tray on 300 mm light-duty runners that rolls 25 cm out towards the room. It holds a full-size wireless keyboard and mouse: pull it out, lift the keyboard onto your lap, park it again. The housing stops 12 cm short of the sofa end so the cables drop past it; its underside is 144.4, 64.4 above the sofa's back.
- **Headroom:** the shelf's underside is at 151.4, and it covers only the 28 cm of the sofa nearest the wall.

7 · The oak joinery

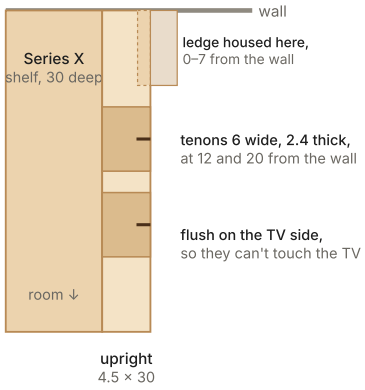
Above the worktop the wall is furniture rather than cladding: two solid oak uprights, shelves carried on joints you can see, and the Card Room Green wall showing between them.

- **Uprights:** solid oak, 4.5 × 30 cm, 159 cm long, each on two 10 mm dowels into the worktop, scribed to the ceiling and held to the back wall with concealed brackets; the left one also screws to the side wall.
- **Wedged through-tenons:** the Series X shelf (top 134) passes through the right upright on two tenons, 6 cm wide and 2.4 thick, 12 and 20 cm from the wall. They're wedged and planed flush: standing proud, they'd touch the TV.
- **Housed and pegged:** the high shelf (two lengths, top 196) and the record ledge (top 155.4) are housed 1.2 cm into the uprights and pinned with 10 mm oak pegs, which show as dark dots.
- **LED grooves:** a routed groove on each upright's inner face holds a 2700K strip that washes the wall either side of the TV.
- **Hidden cables:** a 2 × 1.5 cm groove down the back of the right upright carries Deco B's 12 V lead and patch lead, the Series X's HDMI and power, and the vinyl lead.
- **Finish:** hardwax oil throughout; lightly limed or smoked oak avoids the bright orange "new oak" look.

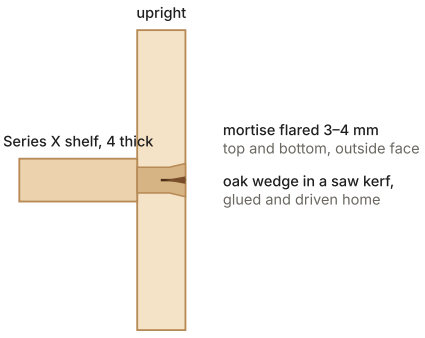
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenons, from above

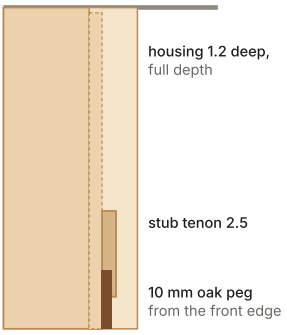


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged



high shelf / ledge
seen from above

• Through-tenons (A, B): the Series X shelf (top 134) passes through the right upright on two tenons, 12 and 20 cm from the wall, wedged and planed flush: proud, they'd touch the TV

• Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.

• Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets. The left one also screws to the side wall, and

Joinery cut list

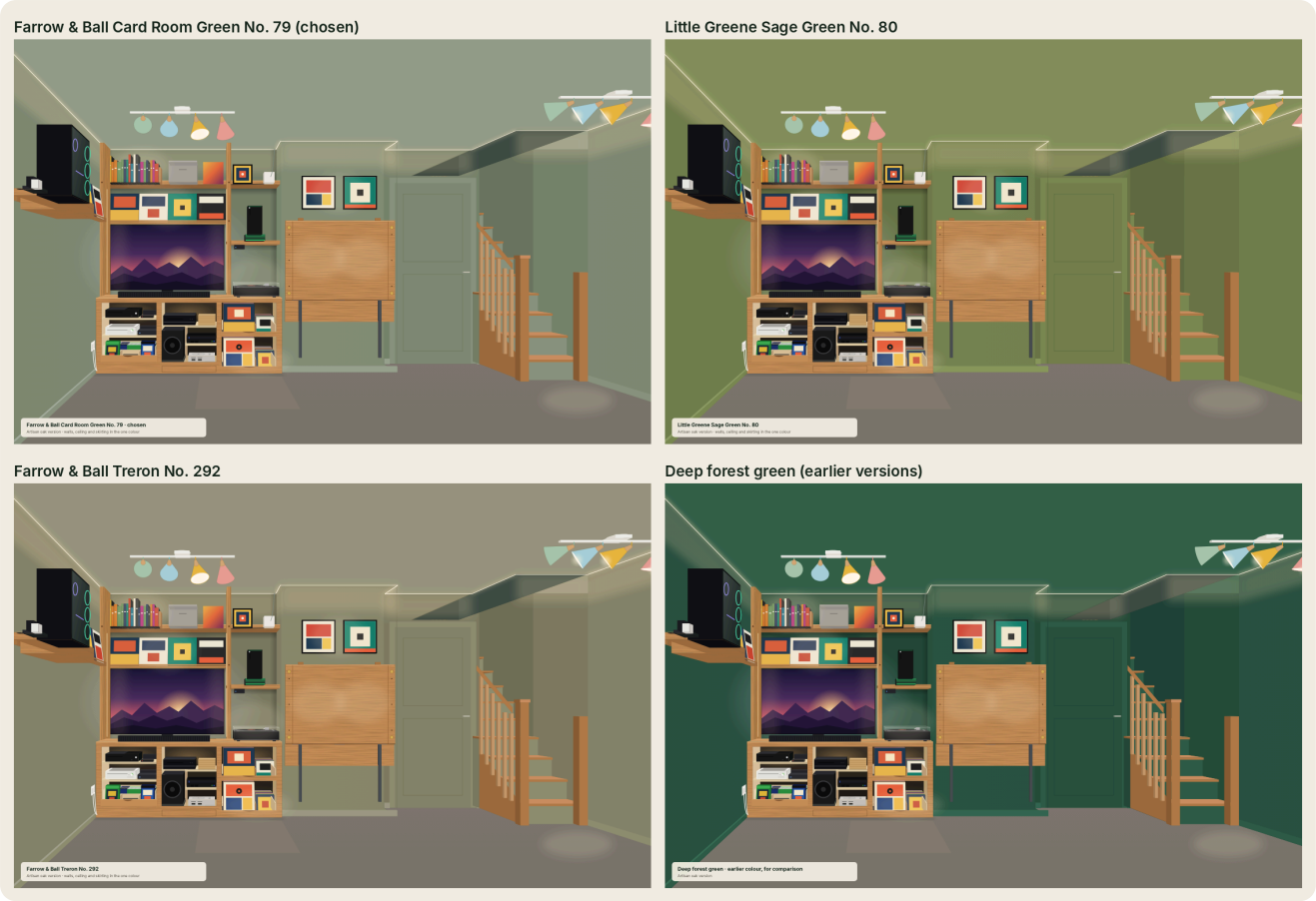
Part	Material	Finished size (cm)	Joints	Qty
Uprights	Solid oak, quarter-sawn if you can	159 × 30 × 4.5 (buy 165 and scribe)	Housings for the high shelf and ledge; mortises for the Series X shelf in the right one; LED groove inside; cable groove at the back of the right one	2
High shelf	Solid oak	128.9 × 30 × 4 and 51.7 × 30 × 4	Housed 1.2 into the uprights, pegged; the short length on a cleat on the reveal	2

Record ledge · wrap ledge	Solid oak	128.9 × 7 × 2.4 · 134 × 7 × 2.4; lips 0.9 × 1.5	Housed and pegged · cleat and two dowels	1 · 1
Series X shelf	Solid oak	55 × 30 × 4 (50.5 + 4.5 through)	Two through-tenons 6 × 2.4, wedged flush; cleat on the reveal	1
PC shelf	Solid oak	82 × 28 × 4	Oak cleat and two concealed brackets	1
Worktop and backboard	One solid oak worktop, 300 × 62	186 × 46 × 3 and 110 × 22 × 3	Chamfered fronts; cut edges sealed	1 board
Face frame	Oak, 20 mm	186 × 73, stiles about 5 wide	Glued and pinned; end stiles scribed	1 set
Wedges · pegs · dowels	Oak · 10 mm oak dowel	6 × 3, tapering · 5 long	Glued	2 + spares · 6 · 8
Keyboard tray	Oak housing sides 7 × 2, ply tray, oak lip	70 × 28 × 7; tray 54 × 26	Housing screwed under the shelf; 300 mm runners	1

8 · Paint and light

Paint: Farrow & Ball Card Room Green No. 79

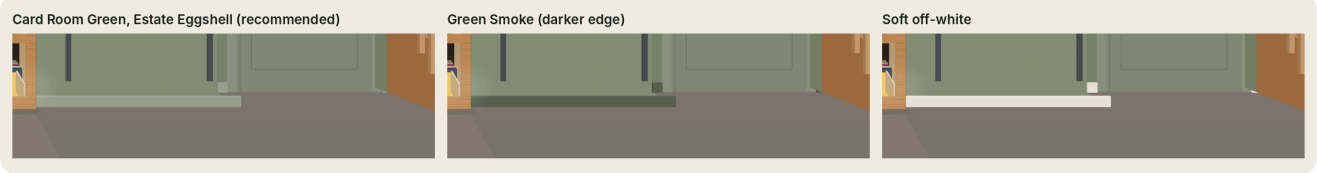
The room is colour-drenched in Card Room Green: walls, ceiling, alcove, chimney breast, skirting, door and frame and the radiator. There's no trunking left to paint. It's a soft grey-green that lets the oak and the records lead.



The wall in Card Room Green (chosen) beside Sage Green, Treron and the earlier deep green. Screen colours are approximate and the renders are lit as in the evening: judge from painted samples.

- **Finishes:** Estate or Modern Emulsion on walls, ceiling, alcove and breast; Estate Eggshell in the same colour on skirting, door, frame and radiator (Modern Eggshell if the skirting takes knocks).
- **Order:** paint before anything is fixed, so nothing needs cutting in round the uprights, shelves or the dining wall.

Skirting: Card Room Green too



Recommended: Card Room Green in Estate Eggshell, which completes the drench and hides scuffs far better than white. Darker option: Green Smoke. Over the white gloss: wash, key, woodwork primer, two coats of eggshell.

Light, in three warm layers

- **The LED strip on the ceiling** (as now): round the room's edge, into the alcove and across the breast; 2700K on a dimmer.
- **Built into the oak:** a 2700K strip in each upright's groove, on a plug-in driver with an inline dimmer, on the 10-gang (no fused spur); a 6500K bias light on the back of the TV from its USB port.
- **A lamp by the sofa (from v6.1):** a floor or table lamp with a 2700K bulb at the sofa's west end. A deep green room needs a low, warm layer as well as the light from above; it's also the one to leave on for a film.
- **Two spot bars** in the style of Dunelm's Wolston (£49 each; 84 × 16 cm, 22.4 cm drop; 4 × E14, 6 W max[14]), parallel to the media wall, on your two existing ceiling points. Bar 1 hangs on the rose straight out from the TV unit, halfway to the sofa. Bar 2 hangs on the point between the door and the west wall, and its heads swing to the table. Each gets an LED-compatible dimmer in place of its switch: like-for-like swaps you do yourselves (step 12), with no new wiring. Bar 1 sits in front of the screen, so tilt its heads up and away: none should be visible in the screen from the sofa.

Head	Bar 1, on the existing rose	Bar 2, over the dining area
Sage	The books and the left of the high shelf	The framed LPs above the table
Pale blue	The record ledge above the TV	The table (the folded top, when it's up)
Mustard	Down onto the coffee table	The table
Dusty pink	The turntable and the Series X	Down onto the way in, by the door

Never point a head at the TV. For six or eight the table stands right under bar 2: point its blue and mustard heads straight down onto it. Use warm-white (2700K) dimmable E14 LED bulbs throughout.

Lighting scenes (v13.7)

Four ways to set the room with the controls it already has: the strip's controller, the inline dimmer on the uprights' LED driver, the two wall dimmers and the lamp's switch. No smart kit needed; if you add smart dimmers later, these are the scenes to save.

Light	Control	TV night	Dinner	Records	Everyday evening
Ceiling LED strip	its own controller	10%	25%	35%	70%
Uprights' LED grooves	the inline dimmer on its driver's lead	15%	30%	60%	50%
TV bias light	comes on with the TV	on	off	off	off
Bar 1, between the TV and the sofa	its wall dimmer	off	15%	60%	70%
Bar 2, by the door	its wall dimmer	10%	80%	15%	40%
Lamp by the sofa	its own switch	on	off	on	on

Levels are roughly how far up each dimmer goes. One dimmer per bar, so its four heads stay aimed as in the table above; a scene changes how bright they are, not where they point.

Lighting scene: TV night

Films and games: the screen is the brightest thing in the room, with a little warm light round it so your eyes don't strain.



Ceiling LED strip
10% · its own controller

Uprights' LED grooves
15% · the inline dimmer on its driver's lead

TV bias light
on · comes on with the TV

Bar 1
off · its wall dimmer

Bar 2
10% · its wall dimmer

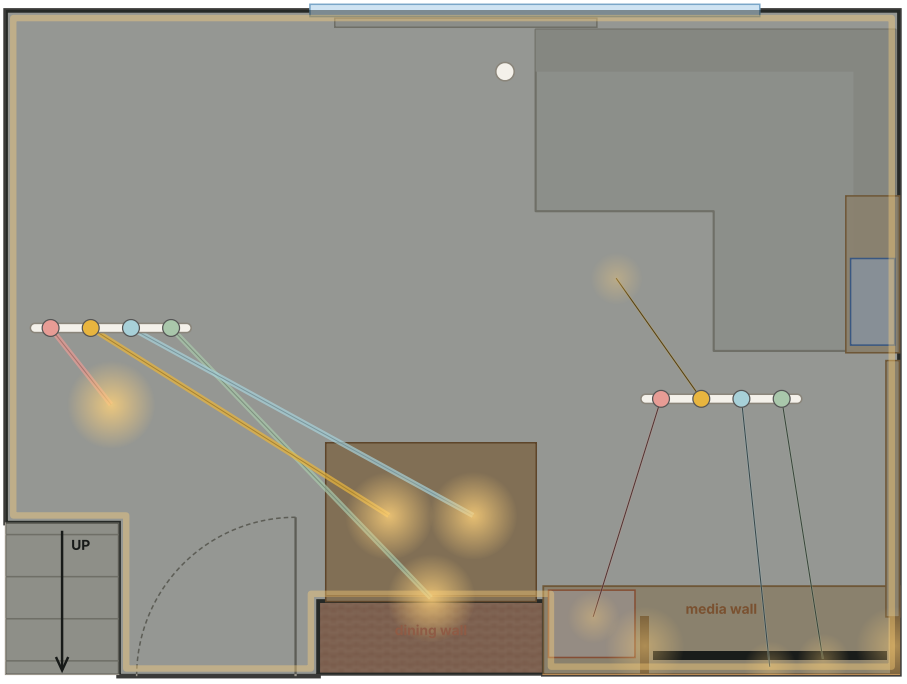
Lamp by the sofa
on · its own switch

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 1 off: its heads hang in front of the screen, and even tilted away they'd catch it.
- The bias light behind the TV and the lamp by the sofa do the work: the lamp is the one to leave on for a film.
- Bar 2 just glowing on the framed LPs, so the far end of the room isn't a black hole.

Lighting scene: Dinner

The table is the bright spot, the media wall steps back.



Ceiling LED strip
25% · its own controller

Uprights' LED grooves
30% · the inline dimmer on its driver's lead

TV bias light
off · comes on with the TV

Bar 1
15% · its wall dimmer

Bar 2
80% · its wall dimmer

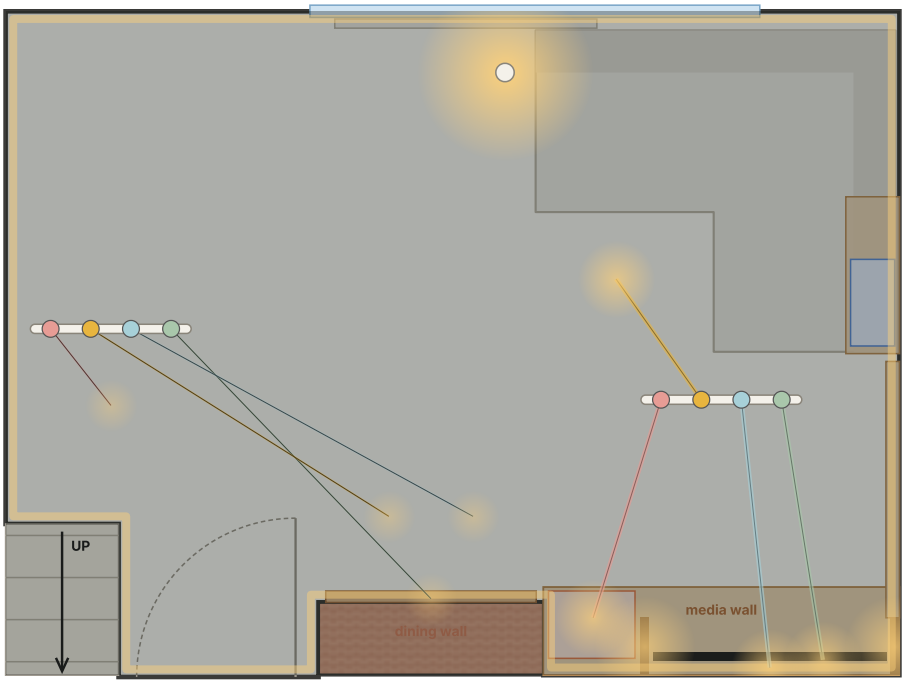
Lamp by the sofa
off · its own switch

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 2 up: for four its heads light the table at the wall; for six, the table stands right under it (point blue and mustard straight down).
- Bar 1 low, so the coffee table end isn't dark; the TV off.
- The uprights low: the shelves read as a backdrop, not a stage.

Lighting scene: Records

Pick a record, read the sleeve: light on the deck, the ledge and the trays.



Ceiling LED strip
35% · its own controller

Uprights' LED grooves
60% · the inline dimmer on its driver's lead

TV bias light
off · comes on with the TV

Bar 1
60% · its wall dimmer

Bar 2
15% · its wall dimmer

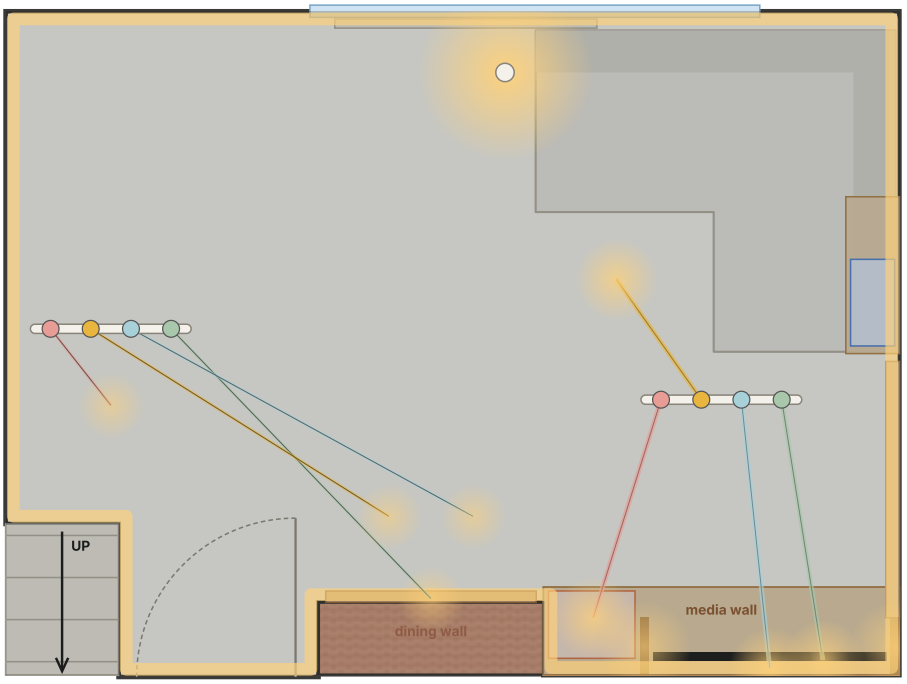
Lamp by the sofa
on · its own switch

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 1 up: its pink head is on the turntable and the Series X, blue on the record ledge.
- The uprights' grooves up, washing the wall either side of the (switched-off) TV and the ledge LPs.
- The lamp on for reading sleeves on the sofa.

Lighting scene: Everyday evening

Everything on at a comfortable level, for tidying, chatting and finding the remote.



Ceiling LED strip
70% · its own controller

Uprights' LED grooves
50% · the inline dimmer on its driver's lead

TV bias light
off · comes on with the TV

Bar 1
70% · its wall dimmer

Bar 2
40% · its wall dimmer

Lamp by the sofa
on · its own switch

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

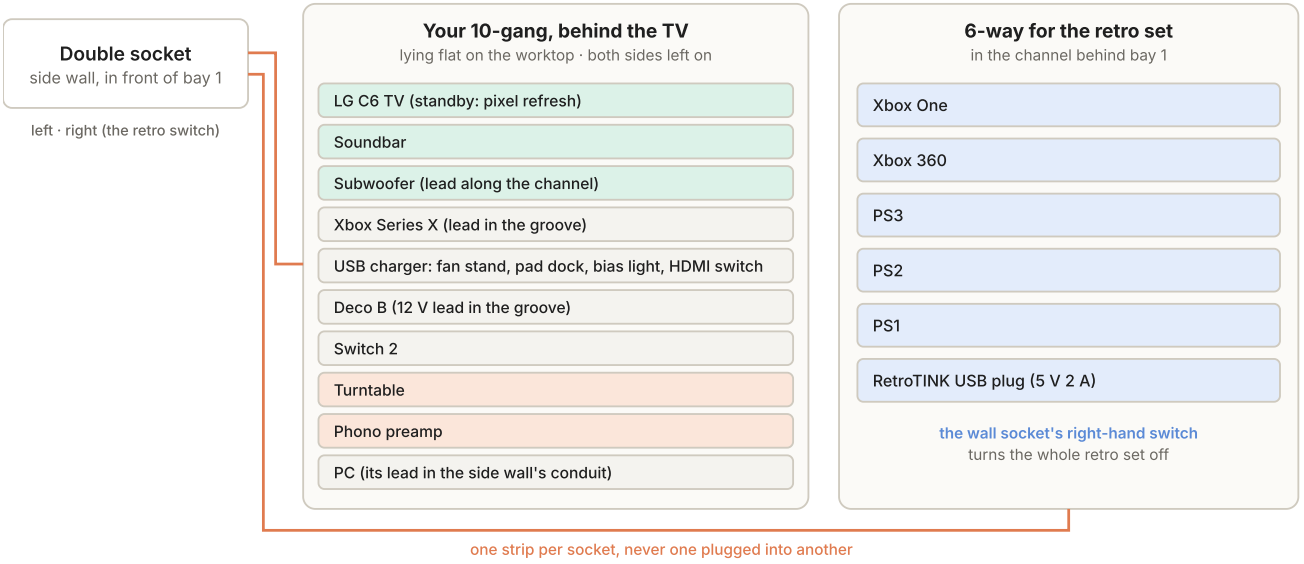
- The ceiling strip does most of it; both bars fill in the shelves and the table.
- Turn the strip down first as the evening goes on: it's the flattest light in the room.

9 · Power, cables and network

Power

Power map: your 10-gang (now with the PC) plus a 6-way for the retro set

Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: you swap it for a double at the same spot (like for like, your own job: step 3).
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- Rear speakers: the socket by the sofa, if there is one. If not, a 12-way surge-protected strip in place of the 10-gang gives them a way.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 1.55 kW with everything on, PC included (850 W at most); the 6-way up to 0.8 kW. Each under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

- **The socket:** the single on the side wall just in front of bay 1, about 30 cm up, becomes a double at the same spot. The left outlet feeds the 10-gang; the right one feeds the retro 6-way, so its own switch is now the retro switch. The 10-gang's switches sit behind the TV, 2 cm from the corner post, where you can't reach them any more.
- **10-gang:** lying flat on the worktop behind the TV, both sides left on: TV, soundbar, sub, Series X, USB charger, Deco B, Switch 2, the deck, the preamp and now the PC. With everything running that's about 1.55 kW (the PC's 850 W at most), under the strip's 13 A (3 kW).
- **Retro 6-way:** in the channel behind bay 1 (bay 2's back is the sub's port): Xbox One, Xbox 360, PS3, PS2, PS1, RetroTINK.
- **OLED care:** leave the TV in standby so it runs its pixel refresh.

Socket by socket

Socket	Feeds	On or off
Side wall, double, left outlet	The 10-gang behind the TV: TV, soundbar, sub, Series X, USB charger (fan stand, pad dock, bias light, HDMI switch), Deco B, Switch 2, the turntable, the preamp, the PC and the uprights' LED driver	Always on: about 1.55 kW with everything running, PC included

Side wall, double, right outlet	The retro 6-way in the channel: Xbox One, Xbox 360, PS3, PS2, PS1, the RetroTINK's USB plug	The wall switch is the retro switch: off between sessions; up to about 0.8 kW with a launch PS3
By the sofa, if there is one	The rear speakers (if there's no socket there, a 12-way strip in place of the 10-gang gives them a way)	On
Ceiling points 1 and 2	The two spot bars, each on an LED-compatible dimmer	Dimmed

One strip per outlet, never one plugged into another, and each under its 13 A (3 kW) limit: about 1.55 kW on the 10-gang and up to 0.8 kW on the retro 6-way, about 10 A together.

Electrics: your own job

You're doing the electrics yourselves. In England, Part P of the Building Regulations makes only three kinds of work notifiable: a new circuit, a replacement consumer unit, and any addition or alteration to a circuit in a special location (round a bath or shower). Replacing an accessory like for like, and adding to an existing circuit anywhere else, isn't notifiable, though it must still be done to the wiring regulations (BS 7671). So the jobs here need no electrician and no building control: the socket swapped for a double, the two dimmers, and the two spot bars in place of the old fittings. The Cat6 and the conduits aren't mains work at all.

The safe routine, every time

1. **Isolate:** switch off that circuit's breaker at the consumer unit (the lighting circuit for the dimmers and bars, the socket circuit for the socket). If the labels aren't clear, switch off the main switch.
2. **Lock off:** fit the MCB lock-off and padlock, keep the key in your pocket, and tape a note on the board.
3. **Prove dead:** check the tester lights on its proving unit, test between every pair of terminals (live–neutral, live–earth, neutral–earth), then check the tester on the proving unit again.
4. **Photograph** the wiring before you disconnect anything, and connect the new one wire for wire: brown (or red) to L, blue (or black) to N, green-and-yellow to earth, sleeved, no copper showing, every screw tight.
5. **Test:** power back on, then the socket tester in both outlets; each dimmer from low to high with no flicker.

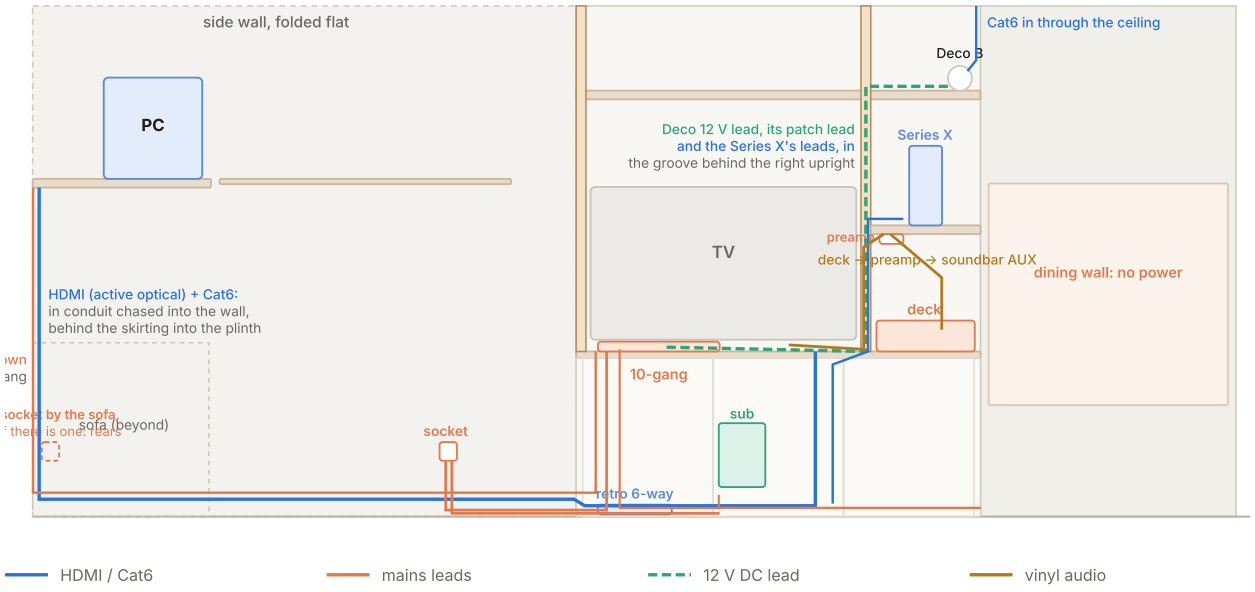
Stop and call an electrician if…

- the tester shows anything live after you've isolated;
- the wiring is old (rubber or fabric sheath, or red and black with no earth), or there's no earth at a ceiling point (the spot bars need one);
- there are more wires than you can match to your photo (two-way switching, a loop in a ceiling rose you can't follow), or scorch marks, or a loose, hot or crumbling fitting;
- the consumer unit has rewireable fuses or no RCD;
- a job grows into a new circuit, consumer unit work, or anything by the bath or shower.

Cable routes

Cable routes: everything ends in the channel behind the base

Front view, schematic, with the side wall folded flat to the left so the PC's route shows; the sub sits in bay 2

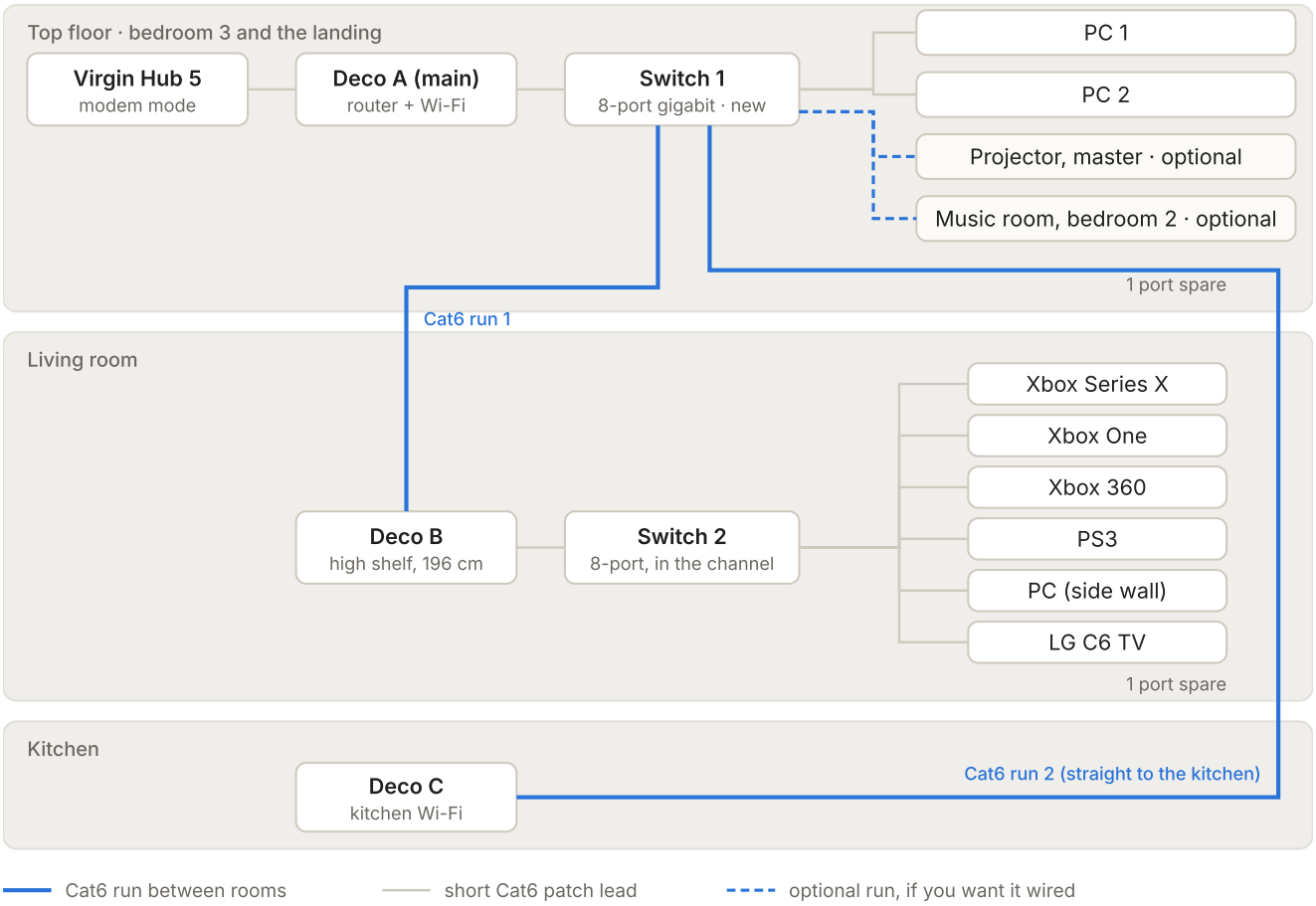


- **The channel:** 4 cm behind the frames, flat two-compartment trunking: mains on one side, HDMI, Cat6 and audio on the other. Switch 2 and the retro 6-way stand on edge in it.
- **The groove behind the right upright:** Deco B's 12 V and patch leads, the Series X's HDMI and power, and the vinyl lead, down to the worktop's slot.
- **The PC:** through the shelf's brush grommet into two conduits chased into the side wall (the HDMI and Cat6 in a 38 mm one, its mains lead in a 25 mm one), behind the skirting and into the plinth notch; the mains lead goes on to the 10-gang.
- **The sub:** its mains lead from the 10-gang drops through the slot behind the TV to the back of bay 2, about 1 m. Keep the channel behind the sub clear: its port needs the air.
- **Airflow:** console bays stay open-backed; the Series X has 15 cm each side and 21 above.

Network

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



- **Bedroom 3:** the Hub 5 (in modem mode), Deco A and Switch 1 sit together with the gaming PCs. Switch 1's eight ports: the uplink, two PCs, the two Cat6 runs down to Deco B and Deco C, and, if you want them, the projector in the master and the music room in bedroom 2.
- **The route down:** across the landing to the airing cupboard, which sits over the alcove's corner, and down through its floor. Run 1 comes through the living room ceiling in the alcove's right corner to Deco B; run 2 follows the right upright's groove down and through the floor to the kitchen, where Deco C goes on the outside of the brick cupboard, up high.
- **Sloped ceilings:** bedroom 2 and the bathroom slope down to the back wall; keep any Deco or desk in bedroom 2 on the high side.

What each part does

Part	Where	Job	Why it's needed
Virgin Hub 5, in modem mode	Bedroom 3	Brings the internet in, nothing else	Two routers fighting (double NAT) breaks Xbox online play; its Wi-Fi switches off
Deco A (main)	Bedroom 3	The router: Wi-Fi, parental controls, HomeShield	HomeShield and parental controls only work when the Deco is the router
Switch 1, 8-port	Bedroom 3	Turns Deco A's spare port into seven	The two PCs, both Cat6 runs down, and the optional projector and music-room runs

Deco B	Living room, high shelf at 196	Wi-Fi for the living room, and the wired point for the wall	Fixes the living room's weak 5 GHz side
Switch 2, 8-port	Standing on edge in the channel behind the base	Turns Deco B's spare port into seven	The Series X, Xbox One, Xbox 360, PS3, PC and TV each get a cable, one port spare
Deco C	Kitchen, high on the outside of the brick cupboard	Wi-Fi for the kitchen	The kitchen gets poor signal on every band today
Cat6 runs 1 and 2	Bedroom 3 → airing cupboard → alcove corner, and on to the kitchen	The wired link between Decos ("Ethernet backhaul")	Without it the Decos relay each other over the 5 GHz band the phones use, losing speed at each hop

Stage 1: the mesh up today, before any cables

1. Unplug everything from the hub. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its settings page moves to 192.168.100.1.
2. Connect a hub port to Deco A and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another port and restart the hub.
3. Plug Switch 1 into Deco A's other port, then the PCs into Switch 1.
4. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi.
5. Check the Xbox's network settings show NAT: Open.

Stage 2: after your Cat6 runs (step 3)

1. Plug run 1 into Deco B's free port (Switch 2 stays in the other) and run 2 into Deco C.
2. In the Deco app, check Deco B and Deco C show as "Ethernet".
3. Plug the Series X, Xbox One, Xbox 360, PS3, PC and TV into Switch 2.
4. Under QoS, give the Series X and the PC high priority; turn on guest Wi-Fi and automatic firmware.

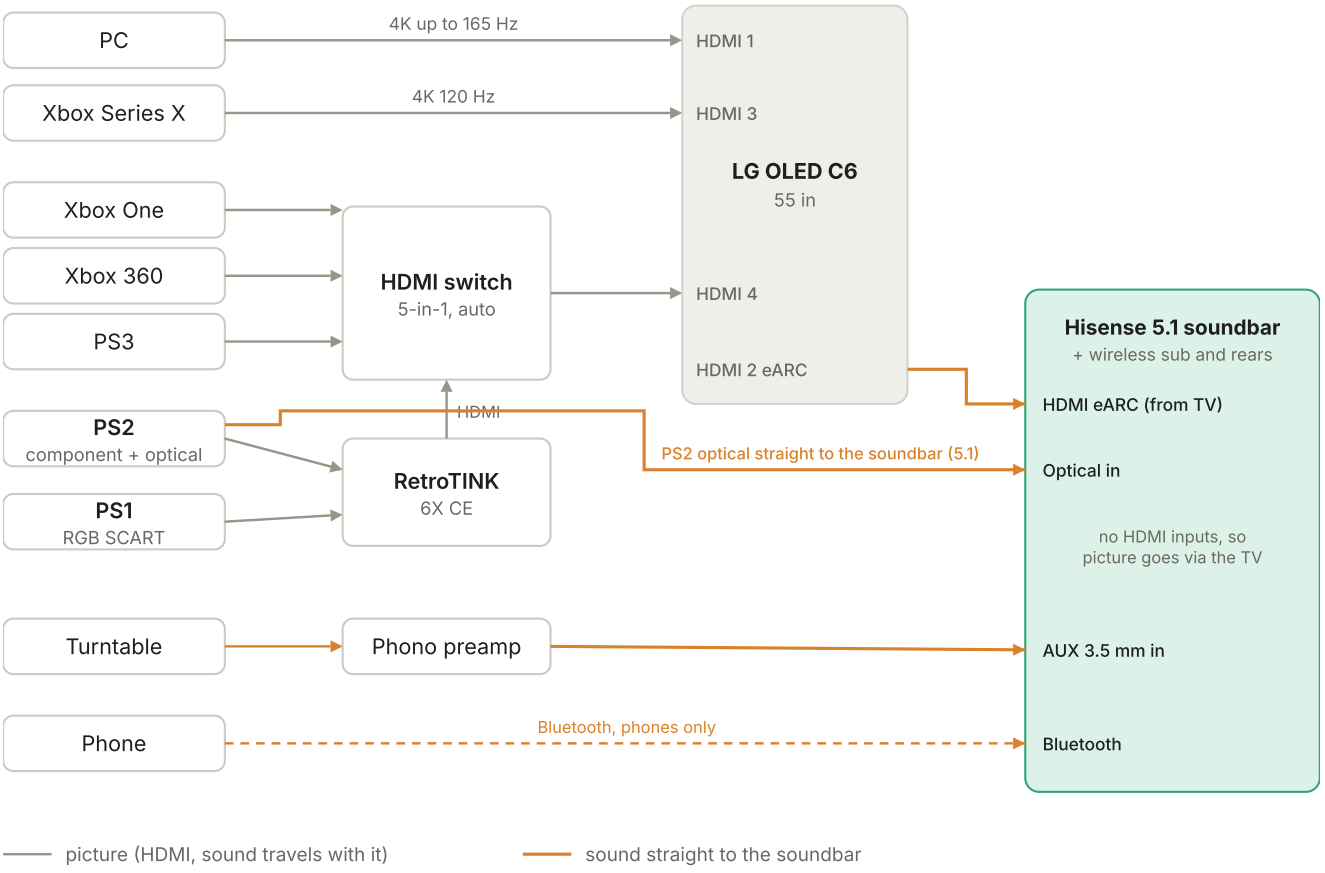
Cabling rules

- Solid pure-copper Cat6, round; not CCA or flat cable for the long runs. Measure the real route and add 2–3 m.
- Cross mains at right angles; never staple through a cable; no kinks; label both ends. Never two cables between the same two devices.
- **Done when:** Deco B and C show "Ethernet", each PC shows 1.0 Gbps, the Xbox shows NAT: Open, and phones hold a strong signal in the kitchen and across the living room.

AV: everything sounds through the Hisense

Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



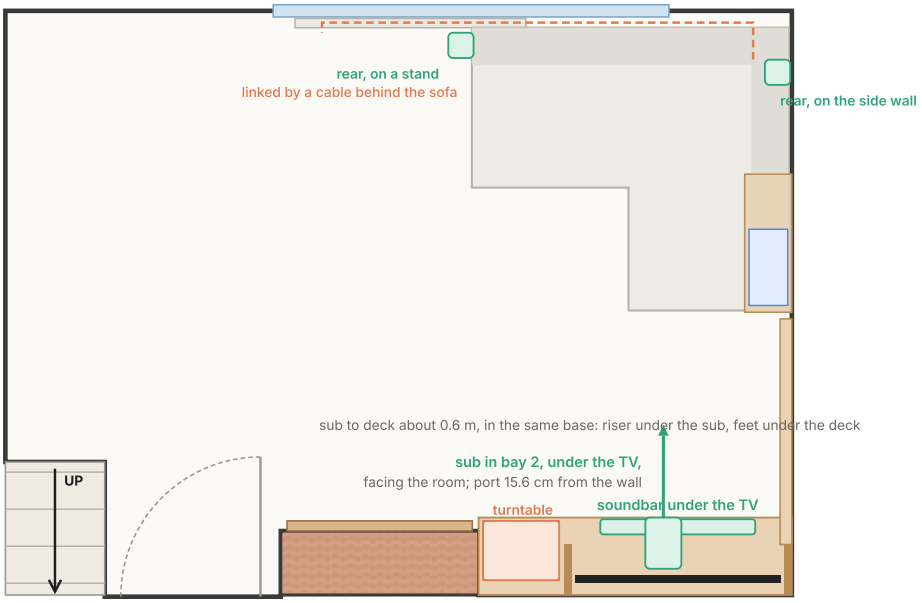
Source	Picture	Sound
PC · Xbox Series X	TV HDMI 1 · HDMI 3	Via the TV over eARC (5.1 or Atmos)
Xbox One, Xbox 360, PS3	HDMI switch → TV HDMI 4	Via the TV over eARC, 5.1
PS2	Component → RetroTINK → switch → TV	Optical straight to the soundbar
PS1	RGB SCART → RetroTINK → switch → TV	Stereo through the RetroTINK and eARC
Turntable · phone	—	Preamp → AUX · Bluetooth

TV settings: rename the inputs (HDMI 4 "Retro"), Deep Colour on HDMI 1, 3 and 4, Game Optimiser on the console inputs, sound out to HDMI eARC. RetroTINK-6X CE: a 5 V 2 A USB plug on the retro 6-way; the PS2's Component Video Out set to Y Cb/Pb Cr/Pr; RGB SCART for the PS1.

Sound layout

Sound: the sub on the front wall with the TV, the rears behind the sofa

To scale, north at the top. The sub is wireless: on the front wall it blends with the soundbar, out of the way of feet



- **Sub:** wireless, in bay 2 under the TV, its driver facing the room and its rear port 15.6 cm from the wall through the open back (15–30 is advised[25]). Next to the soundbar, its bass blends with the bar instead of coming from behind you, and it's off the floor where feet and drinks are. If the bass booms, turn it down rather than moving it.
- **Rears:** one on the side wall at seated ear height behind the sofa's corner, one on a stand past the sofa's west end; linked by a cable behind the sofa, powered from the socket by the sofa if there is one (if not, from a 12-way strip in place of the 10-gang).

10 · Build guide, for a first-time builder

This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Each step carries a coloured label for its part of the house (Prep, Network, Power & cables, Paint, Media wall, Dining table, Lighting, AV), and so does each phase of the buy list. You're doing the electrics yourselves: steps 3 and 12 have the safe routine and the point where you stop and call an electrician. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **The electrics are yours, within limits.** Swapping the socket for a double, the two dimmers and the two ceiling fittings are like-for-like jobs, not notifiable under Part P in England, as long as they meet the wiring regulations. Isolate, lock off and prove dead every time. Anything new (a new circuit, the consumer unit, work by a bath or shower) is an electrician's, and so is anything on the stop list in section 9.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15
Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70

Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
ELECTRICS (YOUR OWN: STEPS 3 AND 12)			
Two-pole voltage tester to GS38, with its proving unit	proving every wire dead before you touch it: non-negotiable	Buy	35–70
MCB lock-off and a small padlock	keeping the circuit off while you work on it	Buy	8–15
Plug-in socket tester	checking the new double socket is wired the right way round	Buy	8–15
Insulated (VDE) screwdrivers, wire strippers, side cutters	the socket, the dimmers and the ceiling fittings	Buy	20–40
Wall chaser with dust extraction (or a bolster and club hammer)	the PC's two conduits in the side wall	Hire	25–45 a day
Cat6 punch-down tool and a cable tester	terminating and checking the Cat6 runs	Buy	20–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.

2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	PREP **Measure and count**	The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest. The floor's level across the alcove, and the chimney breast's flatness with a long level. The door's swing, and the sofa: overall width, depth with the return, back height. Count the LPs, singles and games (PS2, Xbox 360, Xbox One), and measure ten cases of each. Make a cardboard template of the alcove at worktop height: walls are rarely square.	Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.	2 h
2	NETWORK **Network, stage 1**	Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi. Speed-test each room, and check the Xbox's network settings show NAT: Open.	Internet in every room, and NAT: Open on the Xbox.	2 h

3	POWER & CABLES NETWORK **Cables and the socket, your own job**	The Cat6: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner for Deco B, one on to the kitchen for Deco C, a spare pulled with each; the optional runs to the master and bedroom 2 if you want them. Low voltage: no rules beyond care. The socket: swap the single on the side wall for a double at the same spot, with the circuit isolated, locked off and proved dead (the safe routine is in this step). Like for like, so not notifiable. The PC's two conduits: chase them into the solid side wall from behind the PC shelf to skirting height and behind the skirting to the base, 38 mm oval for the HDMI and Cat6, 25 mm for the PC's mains lead, a draw wire in each. Make good. Stop and call an electrician if anything on the stop list turns up (old wiring, no earth, a reading where there should be none, anything you can't match). Before you fill a chase, photograph it with a tape measure in shot, and mark every route on the plan.	The socket tester shows correct wiring on both outlets, both Cat6 runs test good end to end, and every route is photographed.	2–3 days
4	PAINT **Paint**	Paint an A4 board in Card Room Green and look at it by day and with the LED strip on. Wash the skirting and door with sugar soap, sand to a key, and prime them. Two coats of emulsion on the walls, ceiling, alcove and chimney breast. Two coats of eggshell in the same green on the skirting, door, frame and radiator. Leave it a week to cure before fixing anything to it.	Two even coats, and no white showing at the edges or behind the radiator.	2–3 days, then a week to cure
5	PREP **Mark the heights**	With the laser level: 73 and 76 for the worktop, 130 and 134 for the Series X shelf, 153 and 155.4 for the ledges, 192 and 196 for the high shelf. The uprights at 0–4.5 and 131–135.5 cm from the side wall. The dining wall's backboard at 51.3–73.3, centred on the breast. Run the cable detector along every line you will drill, and mark anything it finds.	Every line level end to end, and the uprights' lines plumb.	1 h
6	MEDIA WALL **Build the base**	Order the oak and ply from the cutting list (below the steps), a week or two ahead: most merchants cut to size for a small charge. Build the 9 cm plinth for bays 1–2, with the open bay for the hidden drawer under bay 2. Level it with shims. Assemble the two BESTÅ frames with their back panels left off and stand them on the plinth. Build bay 3 from 18 mm ply, 60 × 40 × 73, and bolt it to bay 2. Screw the 4 cm spacer blocks to the wall, push the base against them and fit the anti-tip brackets. Lay the flat two-compartment trunking in the channel behind (mains one side, data the other) and run the cables in.	The tops level within 1 mm across 186 cm, and nothing rocks.	1 day
7	MEDIA WALL **Fit the worktop**	Cut the worktop and the dining wall's backboard from the one 300 × 62 oak board, using the template from step 1. Scribe the ends to the walls; cut the TV slot over the channel and the grommet hole behind the deck. Oil it on every face, undersides and cut ends included. Screw it up through the frame tops in slotted holes, so the oak can move.	Tight to both walls, and the slot sits over the channel.	½ day

8	MEDIA WALL **Fit out the base**	Shelves on their pins at the fit-proof heights for the consoles. The game drawer in bay 1: runners on the frame's sides, then the box (cut list in the Media wall tab). The record trays in bay 3: runners at 2.3 and 37.0 cm, then each tray. The hidden plinth drawer under bay 2. Bay 2: the divider and the part shelf, then the sub on its riser, its front flush with the face frame.	Every drawer runs smoothly, and the two trays clear each other by 2 cm.	1–2 days
9	MEDIA WALL **Uprights and shelves**	On the bench: cut the housings and the mortises in both uprights, and the LED groove and cable groove. Dry-fit each upright with its shelves before any glue. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the concealed brackets to the back wall. Fit the high shelf and the ledge, then the Series X shelf: wedge its tenons, and peg the housings.	Uprights plumb, shelves level, and the TV slides in with 2 cm each side.	2–3 days
10	MEDIA WALL **Side wall and PC shelf**	Screw the oak cleat for the wrap ledge and the PC shelf to the side wall. Dowel the wrap ledge to the left upright's front edge. Fit the concealed PC shelf brackets, then the shelf. Screw the keyboard tray's housing under it and fit its runners. Fit the brush grommet and pull the PC's leads down the two chased conduits with the draw wires: data in the big one, its mains lead in the small one.	One level line with the ledge over the TV, and the PC shelf holds your weight leaning on it.	1 day
11	DINING TABLE **The dining wall and the leaf's leg case**	Screw the backboard to the breast, plugs every 20 cm. Fit the two Hebgo brackets on its face, 80 cm apart, arm tops level with its top. Screw the lift-off hinges to the backboard's top edge and the table's back edge, all the same hand. On the bench: oil the table's underside and set the four brass inserts for the legs. Hang the table and fit the two turn buttons above it. The leaf's leg case: glue and screw the slim oak frame under the leaf, fit the eight spring clips, and clip all eight legs in.	It folds up and down without binding, both brackets click and lock, and it doesn't move when you lean on it; all eight legs clip into the leaf and nothing rattles when you carry it.	1 day
12	LIGHTING **Lighting: dimmers and spot bars, your own job**	Isolate the lighting circuit at the consumer unit, lock it off and prove it dead at each switch and each ceiling point. Swap both light switches for the LED-compatible dimmers, wire for wire, and take down the two old fittings. Hang bar 1 on the rose between the TV and the sofa and bar 2 on the point by the door, earth first; fit the 2700K bulbs and aim the heads (the table in the Paint and light tab). The LED strips in the uprights' grooves on their plug-in driver (on the 10-gang), the ceiling strip through its notches, the bias light on the TV, the lamp by the sofa. Power back on: each dimmer runs smoothly from low to high with no flicker or buzz. Then try the four lighting scenes (Paint and light).	No flicker at any dimmer setting, and from the sofa no spot head shows in the screen or shines in your eyes.	1 day

13	AV NETWORK **Connect and finish**	TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK). The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.	Everything on the network's "done when" list, and every source plays through the soundbar.	1 day

Roughly 14–18 working days spread over a few weekends, the electrics and the Cat6 included, plus the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the trays before the anti-tip brackets go in.** A full tray weighs about 32 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

The cutting list

Every made part in one list, grouped by what it's cut from, for the timber merchant. Sizes are finished sizes in cm, length × width × thickness. The cut lists in the other sections say how each part joins; this list is for ordering.

Solid oak, hardwax oil

Ask for it planed all round; scribe the uprights and worktop on site

Part	Size (cm)	Qty	Notes	Section
Uprights	159 × 30 × 4.5	2	Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench	Media wall
High shelf, long length	128.9 × 30 × 4	1	Housed 1.2 into both uprights and pegged	Media wall

High shelf, short length	51.7 × 30 × 4	1	Housed into the right upright; cleat on the reveal	Media wall
Series X shelf	55 × 30 × 4	1	Two through-tenons 6 × 2.4, wedged flush	Media wall
Record ledge · wrap ledge	128.9 × 7 × 2.4 · 134 × 7 × 2.4	1 · 1	Housed and pegged · on a cleat, dowelled to the left upright	Media wall
Ledge lips	128.9 × 1.5 × 0.9 · 134 × 1.5 × 0.9	1 · 1	Glued along the ledges' front edges	Media wall
PC shelf	82 × 28 × 4	1	Oak cleat and two concealed brackets	Media wall
Keyboard tray housing	70 × 28 × 7, sides 7 × 2	1	Screwed under the PC shelf	Media wall
Worktop	186 × 46 × 3	1	From one 300 × 62 full-stave worktop; chamfered front	Media wall
Dining backboard	110 × 22 × 3	1	From the same worktop's offcut	Dining table
Face frame	186 × 73 overall, 2 thick, stiles about 5 wide	1 set	End stiles scribed to the walls	Media wall
Table top, centre	101 × 80 × 2.7	1	Edge-glued staves: 96 between the shoulders, plus a 2.5 tongue each end	Dining table

Breadboard ends	80 × 7 × 2.7	2	Pegged with three 10 mm pegs each, in slotted holes	Dining table
Leaf	60 × 80 × 2.7	1	Two alignment dowels and two table clips	Dining table
Leaf frame, long rails	77 × 5.5 × 1.8	2	Under the leaf's joint edge and west end, 3 in; glued along the middle 10 cm only, screwed in slotted holes elsewhere (v13.6)	Dining table
Leaf frame, side rails	50.4 × 5.5 × 1.8	2	Between the long rails, 1.5 in from the leaf's sides; one has eight 10 mm holes for the legs' bolts, 6 apart (v13.6)	Dining table
Legs	71 × 4.5, tapering to 3	8	Tapered on the two inside faces; an M8 hanger bolt in each top; they store in the leaf's frame	Dining table
Turn buttons	6 × 2.5 × 2.2	2	On a single screw each	Dining table
Drawer fronts	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	1 · 1	Finger pull cut underneath	Media wall

Record tray lips	about 56 × 5 × 1.5	2	Cut to the face frame's opening, less 3 mm	Media wall
Record tray dividers	38.2 × 8 × 0.6	2	Between the LP and singles troughs	Media wall
Follower blocks · Xbox One lift block	19 × 12 × 2 · 15 × 12 × 2	2 · 1	Offcuts, felted underneath	Media wall
Wedges · pegs · dowels	6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel	2 + spares · 6 + 6 · 8	Glued	Media wall

Birch ply, 18 mm

Part	Size (cm)	Qty	Notes	Section
Bay 3 sides	73 × 40	2	The record carcass, standing on the floor	Media wall
Bay 3 top and bottom	56.4 × 40	2	Between the sides; back left open to the channel	Media wall
Plinth rails	120 × 9	2	Front and back of the ladder frame under bays 1–2	Media wall
Plinth cross-pieces	34.4 × 9	3	At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer	Media wall

Birch ply, 12 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer sides	35 × 15.5	2		Media wall
Game drawer back · front panel	51.9 × 15.5 · 51.9 × 6	1 · 1		Media wall
Record tray bases	53.9 × 40	2	The sides and ends stand on them	Media wall
Plinth drawer sides · ends	34 × 4.4 · 49.6 × 4.4	2 · 2	A 52 × 34 × 5 box	Media wall
Bay 2 divider · part shelf	35.3 × 38 · 32.2 × 36	1 · 1	The sub's column on the left; the PS2 over the PS1 on the right	Media wall
Keyboard tray	54 × 26	1	Oak lip on its front edge	Media wall

Birch ply, 9 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer base	52.5 × 34.4	1	In a 6 mm groove	Media wall

Record tray sides · ends	40 × 8 · 52.1 × 8	4 · 4	Two of each per tray; new in v13.3: sides set at 8 cm tall, room for the runners	Media wall

Birch ply, 6 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer cross divider	51.9 × 9	1	13 cm back from the front	Media wall
Game drawer bin walls · bin floors	12.6 × 9 · 16 × 12.6	2 · 2	Floors raked so the cases lean about 13°	Media wall
Plinth drawer base	52 × 34	1	Screwed under the box	Media wall

Hardware

Runners are sold in pairs

Part	Size (cm)	Qty	Notes	Section
Record tray runners	400 mm full extension, 68–70 kg a pair	2 pairs	Screwed straight into bay 3's sides	Media wall
Game drawer runners	350 mm full extension, soft-close	1 pair	Accuride DZ3832EC or Screwfix 964VN	Media wall
Plinth drawer and keyboard tray runners	300 mm full extension · 300 mm light duty	1 pair · 1 pair		Media wall
Häfele Hebgo folding brackets	287.43.428, 580 arm	2	500 kg a pair	Dining table
Lift-off hinges	Union PowerLoad 100 × 88, all the same hand	3		Dining table
Brass M8 threaded inserts	for the legs	8	Four in the top's breadboard ends, four in the leaf	Dining table
M8 hanger bolts	M8, about 50 long	8	One in each leg's top, 1.5 left showing	Dining table
Spring clips for the stored legs	RS PRO 45 mm, 42.1 × 56 × 12.7 mm	8	Screwed under the leaf, 8 cm up from each stored leg's foot	Dining table
Sub isolation riser	dense foam or sorbothane, about 22 × 28 × 3	1	Under the sub in bay 2	Media wall
Concealed brackets	for the uprights; two for the PC shelf, 40 kg+ a pair	1 set		Media wall

11 · Buy list and tracker

Every purchase and job, in build order, comes to about £2,764–5,317, with no electrician on top: the electrics are your own job, and the tools for them are on the list. Each phase carries its part's coloured label. Estimates are approximate UK prices; changed and new items are marked. Tick items off and write in what you paid.

Phase	Estimate (£)	Biggest item
NETWORK 1 Network, now	51–70	two 8-port switches
POWER & CABLES NETWORK 2 Cables and power	235–535	the electrical safety kit and Cat6
PAINT 3 Paint	220–540	paint
MEDIA WALL 4 Base unit	412–792	record trays and the BESTÅ frames
MEDIA WALL 5 Oak joinery	715–1,315	the solid oak worktop
DINING TABLE 6 Dining wall	543–985	the oak table top
LIGHTING 7 Lighting	220–470	two spot bars
AV 8 TV, consoles and AV	288–420	RetroTINK-6X CE
AV 9 Vinyl kit	80–190	phono preamp
AV NETWORK 10 Finishing	—	—
Total	**2,764–5,317**	**no electrician: the electrics are your own job (section 9)**

Already owned: Virgin Hub 5, three Deco X1500s, two TL-SG105S switches (both become spares), LG OLED C6 55" and its stand, a 10-gang, the Hisense AX5100G 5.1 soundbar, the Series X fan stand, the turntable, the L-sofa and all the consoles.

Status	Item	Qty	Estimate (£)	Paid (£)
NETWORK	**1 NETWORK, NOW**			
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
POWER & CABLES NETWORK	**2 CABLES AND POWER**			
☐ To do	Cat6, your own job: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner (Deco B) and one on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2 **(changed: no electrician)**			
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	Cat6 tools: a punch-down tool, RJ45 keystones and two surface boxes (or plugs and a crimper), a cable tester, a draw-rod set **(new: you run the Cat6 yourselves)**	1 set	30–60	
☐ To do	Swap the single socket on the side wall for a double at the same spot: like for like, not notifiable; step 3 has the safe way to do it **(changed: your own job)**			
☐ To buy	A double socket in the same finish, and a deeper back box if the old one is shallow **(new)**	1	5–15	
☐ To buy	Electrical safety kit: a two-pole voltage tester to GS38 with its proving unit, an MCB lock-off and padlock, a plug-in socket tester **(new: for doing it yourselves)**	1 set	45–90	
☐ To buy	Insulated (VDE) screwdrivers, wire strippers and side cutters **(new)**	1 set	20–40	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To do	Chase two conduits into the solid side wall for the PC, from behind the PC shelf down to skirting height and behind the skirting to the base's plinth: 38 mm oval for the HDMI and Cat6, 25 mm oval for the PC's mains lead; make good and paint **(changed: your own job; the PC now runs from the 10-gang)**			
☐ To buy	Conduit: 38 mm and 25 mm oval, about 4 m each, two draw wires, one oak-coloured brush grommet; a 5 m C13 lead with a rewireable 13 A plug for the PC **(changed)**	1	15–35	
☐ To buy	For the chase: a wall chaser with dust extraction for a day (or a bolster chisel and club hammer), one-coat filler **(new)**	1	20–45	
☐ To buy	PC tidy kit: braided sleeve, shelf-notch grommet	1	5–15	
☐ To buy	Only if there's no socket by the sofa: a 12-way surge-protected strip in place of the 10-gang, so the rear speakers get a way **(new)**	1	0–60	
PAINT	**3 PAINT**			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door and frame; primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
● MEDIA WALL	**4 BASE UNIT**			
☑ Done	Check the alcove back wall for fixings (solid)			
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☑ Done	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid (you confirmed) **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners **(changed: no push-to-open at floor level, as v6.1 said)**	1	30–50	
☐ To buy	Bay 3 carcass for the records: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts **(changed: holds the trays)**	1	45–80	
☐ To buy	Two face-out record tray kits: 12 and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	2	150–240	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	The sub in bay 2: an isolation riser for it (dense foam or sorbothane, about 22 × 28 × 3), 12 mm birch ply for the divider (35.3 × 38) and the part shelf (32.2 × 36), four shelf pins, felt pads **(new in v13.5: the sub moves into the media wall)**	1	25–45	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
● MEDIA WALL	**5 OAK JOINERY**			
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, wrap ledge 134 × 7 × 2.4, PC shelf 82 × 28 × 4, lips **(changed)**	1 set	200–340	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners **(new: back from v6.1)**	1	25–45	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
DINING TABLE	**6 DINING WALL**			
☐ To buy	Solid oak table top 110 × 80 × 2.7 with pegged breadboard ends, and a 60 × 80 × 2.7 leaf (edge-glued staves from a timber merchant, who can also radius the corners) **(new)**	1 set	220–400	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	
☐ To buy	Eight tapered oak legs, 71 cm, 4.5 tapering to 3, each with an M8 hanger bolt (made from the cutting list, or bought ready-tapered): four for the top, four for the leaf **(changed in v13.6: oak only, so they store in the leaf)**	8	70–160	
☐ To buy	The leaf's leg case: oak for its frame (two rails 77 × 5.5 × 1.8, two 50.4 × 5.5 × 1.8), eight RS PRO 45 mm spring clips, felt **(new in v13.6)**	1	25–45	
☐ To buy	Two oak turn buttons, two leaf-alignment dowels and two table clips for the leaf, felt pads **(new)**	1	10–20	
☐ To buy	Four folding chairs, IKEA TERJE or similar (skip if yours fit: seat about 45–48) **(new)**	4	100–160	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	
☐ To buy	Eight M8 brass threaded inserts for the legs, four in the top's breadboard ends and four in the leaf, set into the underside instead of plates **(changed in v13.5: the leaf's four)**	8	12–20	
☐ To do	Fix the backboard to the breast, hang the brackets and the top, check it folds and locks; set the legs to the right height			
LIGHTING	**7 LIGHTING**			
☐ To do	Swap both ceiling fittings for the spot bars and both light switches for LED-compatible dimmers: like for like, your own job; step 12 has the safe way **(changed: no electrician)**			
☐ To buy	Two LED-compatible dimmers (trailing edge), in your switches' finish, rated for the bars' LED load and your back boxes' depth **(new: on the buy list now)**	2	30–60	
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, a plug-in driver with an inline dimmer (on the 10-gang, so no fused spur); 6500K bias light for the TV **(changed)**	1	40–90	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars (£49 each), plus eight warm-white (2700K) dimmable E14 LED bulbs, 6 W max **(changed: two)**	2	120–160	
☐ To buy	A floor or table lamp by the sofa with a 2700K warm LED bulb: the low layer a green room needs **(new: from v6.1)**	1	30–80	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
AV	**8 TV, CONSOLES AND AV**			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
AV **9 VINYL KIT**				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
AV NETWORK **10 FINISHING**				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			

12 · Still to confirm, later ideas, sources

Still to confirm

- ☐ Where the dining table goes: on the chimney breast, or under your shelf by the stairs (both are drawn and checked).
- ☐ The stairs wall: solid, or plasterboard on studs? It decides the bracket fixings if the table goes there.
- ☐ The floor lamp stays (we read "only 100%" as yes) and can move: by the window, at the sofa's end or beside the breast?
- ☐ The electrics: you're swapping the light fittings, the dimmers and the socket yourselves (as planned), or an electrician for the fittings only?
- ☐ The sofa's return: how far it runs along the side wall (the long side is about 188 from your photo; the return is drawn at 168).
- ☐ The stairs wall's length and the shelf's ends and height (the photo gives about 257–268 cm, shelf 41 from the stairs, 16 from the corner, top about 200).
- ☐ The turntable's model (the zone takes up to 45.2 × 35.2 cm) and whether it has a PHONO/LINE switch.
- ☐ The Series X fan stand's model: the shelf takes a footprint up to about 30 cm wide.
- ☐ Dining chairs: use ones you have (seat 45–48), or four folding ones? And where they'll be kept: the leaf, legs inside, goes with them.
- ☐ The 23 × 22 cm box in the window's west corner.
- ☐ Records: roughly how many LPs and singles (the wall holds about 188 and 240), and are most in thick outer sleeves?
- ☐ Games: roughly how many PS2, Xbox 360 and Xbox One games? The drawer holds about 48.
- ☐ Wired ports for the projector (master) and the music room (bedroom 2)?
- ☐ The 10-gang: is it surge-protected, where are its two switches, and how long is it? It now feeds the PC too.
- ☐ Console model numbers, from the labels underneath, to tighten the row heights; the plan assumes the largest version of each.
- ☐ The pendant point by the stairs wall (the photo gives about 55 cm out, a quarter of the way along the shelf).
- ☐ The RetroTINK-6X CE's size when it arrives (bay 1 allows 23 × 23 × 8).

Easiest way to answer: the viewer page has a box for each of these and takes photos; or reply in a message, a line each.

Agreed so far

- ✓ From your v13.6 message: the legs store as part of the table (in the leaf's frame); the electrics are your own job; each step labelled with its part.
- ✓ From your v13 message: less cladding (done: no slats, spine, sub box or reeding; cables chased into the wall); the stairs go up; the drawer layouts drawn.
- ✓ From your answers (v12.1): the sofa's long side ends in line with the breast's reveal; the door hinges on the breast side and opens in; the side wall and breast are solid (plugs and screws, no stud fixings); the PC is an average gaming case (allowed for at 23 × 45.3 × 46.6).
- ✓ Artisan oak, Card Room Green with matching eggshell skirting, every record face-out, the game drawer as designed.
- ✓ The TV 25.25 cm nearer the side wall, the turntable in the alcove, the Series X above it, the records below.

- ✔ The ledge wrapping round onto the side wall, with the PC above the sofa.
- ✔ A fold-down oak dining table on the chimney breast; the spot bars on the existing ceiling points; the sub back in the media wall, in bay 2 (v13.5).

Later ideas

- **Dave and Emma's watch log:** a shared film and TV log, like a Spotify Blend for what you watch: log each film or episode, mark whether you watched it together, rate it, and later get suggestions for what to watch next. To be shaped up separately.

Sources

Numbered sources are the item sizes used in the fit proof (section 3).

1. Your hand-measured sketch of the living room, in mm (alcove 1860 × 380, chimney breast 1175, ceiling 2350)
2. LG: OLED55C6 specifications: 1222 × 703 × 45.1 mm without stand, 1222 × 757 × 230 mm with stand, stand 470 × 230 mm, 14.1/15.9 kg, VESA 300 × 200 (lg.com, OLED55C6 product pages)
3. Trusted Reviews and Expert Reviews, Hisense AX5100G reviews: soundbar 920 × 60 × 90 mm, subwoofer 214 × 294 × 304 mm (W × H × D) with a rear-firing port, rear speakers 90 × 144 × 109 mm
4. Wikipedia, Xbox Series X: 151 × 151 × 301 mm; Pure Xbox (Nov 2020) quoting Microsoft: at least 4 cm clear above the console on an open stand, 10–15 cm recommended on all sides
5. IKEA: BESTÅ frame 60 × 40 × 64 cm (10247379); BESTÅ shelf 56 × 36 cm, 20 kg maximum load
6. TP-Link: TL-SG108 specifications, 158 × 101 × 25 mm (tp-link.com)
7. Laptop Outlet: TP-Link Deco X1500 3-pack listing, 110 × 110 × 114 mm per unit
8. Audio-Technica: AT-LP120XUSB specifications, 452 × 352 × 141.6 mm, 8 kg without dust cover (audio-technica.com)
9. RetroTINK: RetroTINK-6X CE shop page (power 5 V 2 A by USB; no size published)
10. UGREEN: 5-in-1 4K60 HDMI switch CM568 (90512), 127 × 59 × 25 mm
11. Original Xbox One power supply: 17 × 7.5 × 4.8 cm (6.69 × 2.96 × 1.88 in), as listed by several replacement-PSU sellers
12. Wikipedia: Comparison of PlayStation consoles (PS1 270 × 188 × 60, PS2 302 × 183 × 78, PS3 325 × 274 × 98 mm); Xbox One (333 × 274 × 79 mm); Neowin: official Xbox 360 dimensions (309 × 258 × 83 mm)
13. Record and case sizes: LP sleeve 31.4 cm (12⅜ in); 7-inch sleeve 18.3–18.4 cm; DVD-style keep case 19 × 13.5 × 1.4 cm; Xbox One case 17 × 13.5 × 1.1–1.2 cm (videogamecasesize.info; Walvis Products; Wikipedia: Keep case)
14. Dunelm: Elements Wolston 4-light spotlight bar, £49: 84 × 16 cm, 22.4 cm drop, shades 15 cm, E14 6 W max (dunelm.com, product 1000189449)
15. Emuca heavy-duty full-extension runners, 400 mm, 68 kg per pair; Accuride DZ3832EC 350 mm, 43 kg; runner fitting allowance 12.5 mm a side
16. Xbox 360 original 203 W power supply: no Microsoft size published; Akyga's 203 W replacement for it (AK-PD-14) is 210 × 77 × 55 mm (b2b.akyga.com), so an allowance of 21 × 10 × 6 cm is used here (measure yours)
17. Turntable Lab: Audio-Technica replacement dust cover for the AT-LP120XUSB (705-140XP-1815), 17.375 × 13.625 × 2.75 in (44.1 × 34.6 × 7.0 cm)
18. Häfele UK, Technology catalogue 2023, p. 7.206: Hebgo heavy-duty folding bracket 287.43.428, arm 580 mm, height 220 mm, 500 kg per pair, locks automatically when raised; up to 800 mm between brackets (Häfele 287.43.419 listing)
19. Screwfix: Union PowerLoad lift-off hinges, 100 × 88 mm, 3-pack (5938T), 120 kg maximum door weight
20. Emuca: table legs Ø60 mm, 710 mm, height adjustable 710–730 mm, 50 kg per leg (set of 4): 710 is the standard dining leg; the plan's oak legs are made to the same 71 cm
21. RS PRO 45 mm steel spring clip (RS 175-3588): 42.1 × 56 × 12.7 mm, packs of 10 (uk.rs-online.com)
22. IKEA: TERJE folding chair, 43 W × 49 D × 78 H cm, seat height 48 cm, tested for 110 kg
23. Book a Builder UK, small dining room ideas: 600 mm of table edge per diner; table edge to a wall 600 mm tight minimum, 750 mm to aim for, 1000–1100 mm to walk behind; dining height about 750 mm
24. Your photo of the sofa (18 Oct 2026): a black jumbo-cord L-sofa, the long side under the window ending in line with the breast's reveal, about 188 cm; the return's 168 is a typical corner-sofa size until measured
25. Klipsch subwoofer placement guidance, as quoted by Cloudspress and TechYorker (2026): leave about 6–12 in (15–30 cm) between a rear-firing port and the nearest obstruction
26. Xbox Support (Microsoft), About the Xbox One power supply (archived): don't block its vents, don't put it in a confined space such as a bookcase, rack or stereo cabinet unless the space is well ventilated, don't stack devices on it, and lay it flat on its four rubber pads; the Xbox 360 brick's advice is the same

27. Xbox Support (Microsoft), Xbox 360 placement tips, as quoted on Gaming Stack Exchange: an open, well-ventilated area, a hard surface, and away from high-vibrating sources such as subwoofers
28. Your two photos of the west end (20 Oct 2026): the 2 m black shelf on the wall beside the stairs, scaled from its 2 m: it starts about 41 cm past the stair box and stops about 16 cm short of the window corner, top about 200; the pendant hangs about 55 cm out from the wall, a quarter of the way along it
29. Corsair 4000D Airflow, a typical mid-tower gaming case: 453 × 230 × 466 mm (TechPowerUp review)
30. Virgin Media: Hub 5 modem mode
31. RetroTINK: Introducing the RetroTINK-6X CE (inputs including TOSLINK, deinterlacing); RetroTINK-2X Pro listed as legacy; RetroTINK-4K CE
32. Expert Reviews: LG C6 OLED review (4 × HDMI 2.1, one eARC, ports on the left); JB Hi-Fi: LG C6 specifications (165 Hz mode for PC sources)
33. Hisense AX5100G spec sheet (inputs, auto power-on); JB Hi-Fi Business: AX5100G (rears linked by cable, wireless to the soundbar)
34. TP-Link: Deco X1500 specifications (two gigabit ports per unit)
35. Glorious Vinyl Vault specifications (80 LPs and 24.5 kg per drawer), a check on the tray capacity
36. Screwfix: Essentials 964VN 350 mm runners; Metrol: Accuride DZ3832EC; IKEA: BESTÅ drawer runners
37. NIST SP 500-252 / CLIR pub121: Care and Handling of CDs and DVDs (store upright, book style)
38. Sony PS3 and Microsoft Xbox 360 S manuals: ventilation; IKEA UK: secure it
39. Ideal Home: colour drenching; The Paint Shed: colour drenching with Farrow & Ball; Livingetc: sage green shades (Card Room Green, Treron, Sage Green, Green Smoke)
40. Pontata: Card Room Green vs Little Greene Sage Green (approximate screen colours)
41. Homebase and NC Tiles: Farrow & Ball Card Room Green No. 79 in Modern Eggshell and Estate Eggshell
42. haart Medway: floor plans and listing photos for the house (Portsmouth Close)
43. ConsoleMods: PS2 Component Video Out setting
44. GOV.UK: Electrical safety, Approved Document P (2013 edition, for use in England): notifiable work is a new circuit, a replacement consumer unit, or an addition or alteration to a circuit in a special location (round a bath or shower); like-for-like replacements and additions to existing circuits elsewhere are not notifiable (gov.uk/government/publications/electrical-safety-approved-document-p)
45. IET Electrical: Part P, England and Wales (notifiable and non-notifiable work); Planning Portal: Electrics, Building Regulations (non-notifiable work must still meet the regulations, BS 7671)

Appendix · Options we set aside

Options we rejected along the way

Option	Why not
Slatted oak cladding: cable spine, PC screen, turntable panel, sub box, reeded table (v12)	You asked for less cladding: the chased conduit and plain oak do the same jobs more quietly
The slat wall alternative (v7–v12)	Cladding over the whole alcove; the oak uprights won
The vinyl cabinet on the chimney breast (v10)	The breast is the only clear wall for a table; the turntable fits in the alcove now
Hub as router, Decos as access points	Loses HomeShield and parental controls
All-wireless mesh · powerline	Speed drops at each hop · slower and less stable than Cat6
Records spine-up in one drawer	Showed only a handful of covers; face-out trays show every one
A green feature wall only · white skirting	The drench ties the room together; white cuts a bright line round it
Bluetooth from the turntable	Compresses the sound; the AUX lead keeps full quality
Folding legs fixed under the table · a box for the legs under the dining wall · the plinth drawer (v13.6)	They'd show on the folded panel and stand it 5 cm off the wall · it sits where the chairs push back with six · too short for 71 cm legs. The leaf carries them instead.
An electrician's visit for the socket, dimmers and fittings (v13.6)	You're doing them yourselves: like-for-like, not notifiable; a new socket behind the sofa and a fused spur are dropped